INCH-POUND

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MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, M692 AND M731
HOUSING, TIMING AND FUZING FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembling and packing for the Housing, Timing and Fuzing (HTF) used in Projectiles, 155 MM, M692/M731 (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards, and handbooks forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, General General Quality Assurance Provisions, General Specification For.
MIL-M-48245	- Molding Compound, Housing and Plug.
MIL-I-48331	- Integrated Circuit, Digital, COMS, Time Base and Special Purpose Function Generator, Monolithic, Silicon.
MIL-B-48343	- Battery, Primary Reserve, Ammonia, 5 Milliampere Hour Capacity
MIL-R-63993	- Resistor, Thick Film, Fixed and Trimmable.
MIL-T-70638	- Test Projectile, 155MM, Non HE, for M692/M731 Test Mines Loading, Assembling, and Packing

STANDARDS

FEDERAL

FED-STD-595	- Color (Requirements for individual color chips).

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-453	- Inspection, Radiographic.
MIL-STD-750	- Test Methods for Semiconductor Devices.
MIL-STD-883	- Test Methods and Procedures for Microelectronics.
MIL-STD-2000	- Soldering Technology, High Quality/High Reliability

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.9)

ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

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PRODUCT AND PACKAGING DRAWINGS

9298319	-	Special Test Projectile for M692/M731, 155MM.
9298399	-	Special Test Mine, Type 1 and 2.
9298450	-	Housing, Timing, and Fuzing.
9313636	-	Box, Fiberboard for Housing, Timing and Fuzing Type 1/Type 2.
9313638	-	Box packing and marking drawing.

(Copies of drawings, publications and other Government documents required by the contractor in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117-73	Standard Method of Salt Spray (Fog) Testing.
ASTM B201-80	Standard Practice for Testing Chromate Coatings on Zinc and Cadmium Surfaces.
ASTM D695-(69)	Compressive properties of Rigid plastics.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.1.4 Order of precedence . In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications and standards.

3.2 Housing, timing and fuzing. The Housing, Timing and Fuzing shall comply with all requirements specified on Dwg. 9298450, all associated drawings, and with all requirements in applicable specifications and standards.

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3.2.1 The definition of "damaged" with respect to the shorting bar shall be that the shorting bar shall not be cracked or broken. Bent, nicked, scratched, or otherwise questionable units shall be verified for continity either by electrical test or by radiographic inspection. If the unit is found to be continuous by these tests, the unit shall be considered acceptable for Special b of Paragraph 4.4.2.23. Units deemed questionable but acceptable to Special b. of Paragraph 4.4.2.23 shall be evaluated per Minor 208 of Paragraph 4.4.2.23.

3.3 S&A assembly.

3.3.1 Presence of sliders and springs in S&A assembly. The S&A assembly shall have the long and short sliders, and the long and short slider springs, present and properly positioned.

3.3.2 Qualification test of electric detonator and lead cup assembly lots. The electric detonator serial lot and lead cup assembly serial lot to be used in the housing lot (see 4.4.1) shall perform satisfactorily as evidenced by high order functioning of the lead cup assembly when subjected to the qualification test specified in 4.5.6.

3.4 Function of sensor assembly. (Dwg. 9298569).

3.4.1 Non-release. The Sensor Assembly, less breakwire retaining ring, cover and sealants, when subjected to an acceleration of 9000 g's minimum, in the direction of deployment, shall not release when tested in accordance with 4.5.7.1.

3.4,2 Release. The Sensor Assembly when subjected to a pressure of 250 + 25 psig for 1 second maximum while at a temperature of 75 ± 10°F shall release the ejection spring, sleeve, cap and bobbin with the thread remaining unbroken and attached to the bobbin and post and with the breakwire maintaining continuity when tested as specified in 4.5.7.2.

3.4.3 Tripline deployment. When the Sensor Assembly functions in 3.4.2, the tripline shall deploy a minimum of 16 feet when tested in accordance with 4.5.7.3.

3.4.4 Breakwire function. When the pressure is released and the line pulled, the breakwire shall break at a force of 140 grams, minimum, and 500 grams, maximum, when tested in accordance with 4.5.7.4.

3.4.5 Post retention and thread breaking strength. Following the breakwire function, the post must be retained within the sensor case and the thread must not break when a 540 gram minimum dislodging force is applied in accordance with 4.5.7.5.

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3.4.6 Molding material intrusion test. After molding into a test brick, the sensor assembly when subject to 250 + 25 psig for 1 second maximum and while at a temperature of 75° + 10°F, shall release the cap, sleeve and bobbin assembly when Tested as specified in 4.5.7.7. The sensors shall meet the sampling requirements as defined in 4.4.4.

3.5 Functioning of electronic circuitry. After molding into the Housing, Timing and Fuzing, the electronic circuitry of the Housing, Timing and Fuzing shall be tested as specified in Appendix A.

3.6 Mine loading. All non-hazardous mines shall be loaded and packed at a Government approved loading facility for final lot acceptance testing.

3.7 Temperature control during molding and cure of housing, timing and fuzing. The Housing, Timing and Fuzing shall not have been subjected to temperatures exceeding 165°F at any time when examined as specified in 4.5.11.

3.8 Soldered electrical connections. Soldered electrical connections on the following assemblies shall comply with MIL-STD-2000 l

Printed Wiring Board Assembly	(Dwg. 9298495)
Preform and Sensors Assembly	(Dwg. 12927051)
Preform Assembly	(Dwg. 12927051)
Housing, Timing and Fuzing	(Dwg. 9298450)
Switch Assembly (Terminals)	(Dwg. 9298512)

3.8.1 Breakwire soldered connections. The breakwire soldered connections on the sensor assembly (dwg. 9298569) shall comply with the requirements of 4.5.7.6 and MIL-STD-2000.

3.9 Void criteria for housing, timing and fuzing. Voids with dimensions larger than 1/8 inch across by 1/16 inch deep are not allowed except within the large cavity below 1/4 inch from the open end where voids with dimensions greater than 1/4 inch across by 1/4 inch deep are not allowed. In addition, voids in the SS&A and penthouse area of ball cavity allowed to maximum dimension of 1/8 inch across by 1/8 inch deep, except the booster pocket must be intact.

Comparison photographs or visual standards may be used to establish visual accept/reject criteria for voids as approved by the Contracting Officer.

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3.10 Color. The color of the molding compound in the housing, timing and fuzing shall be within the range of colors bounded by color numbers 34079 and 34102, inclusive, of FED-STD-595.

3.11 Mine functions.

3.11.1 Anti-disturbance functioning. The mine shall function in the anti-disturbance mode when the mine is disturbed by two (2) different movements to two (2) different adjacent faces.

3.11.2 Breakwire functioning. The mine shall function in the breakwire mode when any deployed tripline is pulled with a minimum force of one hundred and forty grams (140 g.).

3.11.3 Self-destruct functioning. The mine shall be capable of self-destructing itself within the required time limits for either a Type I or Type II device.

3.11.4 Early functioning. The mine shall not self-destruct prior to activation by any internal mode within the minimum self-destruct time period.

3.12 Tripline deployment. The mine after ground impact shall deploy a minimum of three (3) triplines with at least one (1) deployed tripline attaining a minimum deployed length of thread of ten (10) feet.

3.13 Duds.

3.13.1 Safe dud. The mine, if not functioned by any external or internal mode, shall not function after the maximum self-destruct time has elapsed.

3.13.2 Hazardous dud. All safe duds when tested as specified in 4.5, shall not be capable of functioning after the maximum self–destruct time has elapsed.

3.14 Test temperature and conditioning. The loaded projectiles, non-hazardous, shall be temperature conditioned for a minimum of twenty-four (24) hours prior to firings. Temperature conditioning and firing temperatures are as follows:

a. Minus twenty-five degrees Fahrenheit, minus five degrees Fahrenheit (-25°F, -5°F).

b. Seventy degrees Fahrenheit plus or minus ten degrees Fahrenheit (+70°F, ±10°F).

One hundred and forty-five degrees Fahrenheit, plus five degrees Fahrenheit (+145°F, +5°F).

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3.15 Test zones. The projectiles, regardless of conditioning temperature, shall be fired either at zone seven (7) or zone three (3) as required per test sequence.

3.16 Compressive strength of molding compound. The molding compound for the Housing, Timing and Fuzing (Dwg. 9298450) shall comply with MIL-M-48245 and shall exhibit a compressive strength of 29,000 to 38,000 psi.

3.17 Qualification testing of Electrical components.. The components of the electronics assembly P/N 9298495 that require qualification by electrical and mechanical testing are as follows:

9298496	Resistor Network
9298507/MIL-I-48331	Integrated Circuit (CMOS)
9298503	Capacitor, Dual
9298509	Capacitor C3
9298658	Capacitor Cl and C2
9298555	Transistors, NPN, Q1, Q2, Q3

Components that were qualified in previous contracts for the Housing, Timing and Fuzing do not require requalification except as specified in paragraph 3.17.4.

3.17.1 Qualification of components. Qualification of the components listed above shall be accomplished by the following:

(1) Electrical and mechanical laboratory tests. - The components shall meet the requirements of the applicable component drawings and specifications and shall be conducted at the contractor's facility. Unless otherwise specified on the applicable component drawing/specification, sampling plans for non-destructive tests shall be 77-1-2 and for destructive tests shall be 20-0-1.

(2) Yuma Proving Ground Ballistic Testing. - The above components must survive a 155MM gun environment (zone 7, M119A2 charge), on a comparative basis with previously qualified components.

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Four (4) rounds for the I.C. and two (2) rounds for the remaining listed components containing 36 mines each shall be fired for evaluation. Each mine shall contain the component/components for qualification. A control group of projectiles equalling the number of qualification rounds shall be fired at the same time and under the same conditions for one-to-one comparison purposes. An optional loading of the projectiles is to load each projectile with an equal number of qualification mines and control mines. The total number of projectiles fired at YPG for qualification shall be equally divided and preconditioned at hot (145°F) and cold (-25°F) temperature.

The rounds, with applicable ballistic charges, will be soaked at temperature for a minimum of 24 hours. They shall then be fired as quickly as possible after being removed from the temperature conditioning chamber. All mines (qualification and control) used for component qualification purposes shall be non-hazardous devices. These mines shall be loaded into projectiles at a Government approved loading facility.

3.17.2 Pass/fail criteria. Pass/fail criteria for qualification components will be as follows:

a. Electrical and mechanical laboratory tests. The componets must pass all tests to the specified sampling plan for qualification as defined on the applicable component drawings and specification, otherwise the qualification sample shall be rejected.

b. Yuma Ballistic Test. Acceptance of the Qualification Sample shall be based upon an overall reliability of 95% at a confidence level of 90% at the component Level. If performance does not achieve this level, a comparison will be made against the qualified components fired at the same time. If the measured reliability of the Qualification Sample equals or exceeds the reliability of the qualified component, the Qualification Sample will be accepted and that vendors part be accepted for use.

c. Yuma Ballistic Test (for I.C. P/N 9298507 only). In addition to the criteria described in 3.17.2 (2) above, the following applies.

(1) A minimum of 200 mines of the 288 ballistically fired must be recovered. 0 to 200 recovered, 100 mines must contain the qualified ADAM I.C. and 100 must contain the Qualification Sample . All lost mines will be considered "NO TEST" devices.

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(2) An I.C. package failure shall be defined as a device (I.C.) failing to function electrically in which no visible I.C. Package damage is observed. This criteria should be interpreted as follows:

(a) An HTF failure attributed to a cracked I.C. in which an arc face crack has propagated through the I.C., shall constitute a NO TEST device but will be recorded and evaluated.

(b) An HTP failure attributed to a cracked I.C. in which there is no evidence of arc face cracking, shall be considered a rejected unit.

(c) An HTF failure attributed to discrete component (i.e., Capacitor, Thickfilm, etc.) shall be considered an acceptable unit with regard to the I.C.

(d) An HTF failure with a multitude of problems (i.e., rejectable Thickfilm Network, etc.) but also a rejectable I.C. shall be considered a failed unit with regard to the I.C.

(3) All HTFs fired shall be retrieved and evaluated for I.C. function. No additional weight shall be placed on any grouping of test devices (i.e., 0 defects allowed for high temp. units). The total defects found out of a total sample shall be the only criteria considered in determining overall final reliability.

3.17.3 Qualification of plating process.

At the components level (Preform, S&A Clip, Electronic Assembly Clip and Sensor Locating Clip).

(1) To qualify the plating process, the Preform, S&A Clip, Electronics Assembly Clip, Sensor Locating Clip shall be subjected to the test criteria for Salt Spray as specified in ASTM B201-80, B117-73. The Type II finish on the preforms and clips shall be capable of withstanding 96 hours of salt testing at a temperature of 35°C + 1.1°C (95°F + 2° - 3°F) and shall not exhibit any signs of corrosion. This test shall be performed prior to the parts being coated with epoxy primer.

b. The testing for the plating process shall be done on a qualification basis only. The qualification sample size shall be 10 parts each (10 preform assemblies and clip assemblies). The parts, after being submitted to salt spray testing, shall not exhibit any form of rust. The sampling plan shall be 10-0-1.

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3.-17.4 Re-qualification. Re-qualification shall be required for all new vendors, major process or design changes by qualified vendors or if a vendor ceases production for over 90 days. The contracting officer shall determine the need for requalification.

3.18 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in paragraphs 4.4.2.1 through 4.4.2.35, as applicable.

3.19 First article inspection. The specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.20 Workmanship.

3.20.1 Workmanship - housing, timing and fuzing. The Housing, Timing and Fuzing Projectile shall be free of dirt, grease, chips, gouges, cracks, flash, fissures, and any other defects which would make the item unsuitable for its intended use.

3.20.2 Workmanship - parts and assemblies. Parts and assemblies shall be tree from chips, dirt, grease, and other foreign matter that may impair improper functioning. Cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated.

3.20.3 Workmanship - packaging, packing and marking. Packaging and packing materials shall not be water-damaged or contain excessive moisture. They shall not be torn, broken or otherwise damaged, and shall be suitable for the intended use. All marking shall be neat, accurate, and sharply defined.

3.20.4 Workmanship allowable rust on components. No rust shall be allowed on S&A clip, electronics assembly clip or sensor locating clips. No rust shall be allowed on the faces of the preform; however, rust is acceptable within 1/4 inch of any edge of the completed Housing, Timing and Fuzing Assembly.

3.20.5 The ‘Visual” method for determining the acceptance of Major 110 of Paragraph 4.4.2.23 shall be that the Housing, Timing and Fuzing shall be examined for presence of green color on the Lead cup.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government . The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified, the provisions of MIL-A-480718 shall apply, and are hereby made a part of this detail specification. Reference shall be made to MIL-STD-109 to define Quality Assurance terms used herein.

4.1.3 Ballistic lot acceptance testing. All ballistic lot acceptance testing shall be performed at a Government testing facility using Government personnel to perform all required testing.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below:

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a. One (1) test bar (0.50 x 0.50 x 1.0 inches or 0.50 inch diameter x 1.0 long) from each molding compound batch (See 4.5.10.1) used to manufacture the Housings in f below. A second test bar shall be made available for use in the event the first bar fails the test in 4.3.2 due to a void. Tolerances on specimen dimensions shall be + 0.01 inches.

b. 20 Barrier Assemblies (Dwg. 9298476).

50 S&A Assemblies (Dwg. 9298464) (less set back pin, setback spring, electric detonator and lead cup assembly) sealing ball, detonator shield, setback cap, sealing washers, sealing balls, and terminal clip.

d. 100 Sensor Assemblies (Dwg. 9298569).

e. 25 Anti-disturbance Switches (Dwg. 9298512).

f. 216 Housing, Timing and Fuzing (Type I or II as specified) (Dwg. 9298450) prior to molding over test pads and adding jumper wires.

g. Certification that each molding compound batch (See 4.5.10.1) used to manufacture the Housing in f complies with the requirements in MIL-M-48245 for compressive modulus, flexural strength, flexural modulus and area under the flexural strength curve.

4.3.2 Inspections to be performed. The First Article parts submitted in accordance with 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the examinations, inspections, and tests specified herein.

a. The test bars for 4.3.1 (a) shall be tested for compressive strength as specified in 4.5.10.2,

b. The twenty (20) barrier assemblies from 4.3.1 (b) shall be subjected to the examinations specified in 4.4.2.2 and shall then be subjected to the shear test specified in 4.5.5.

c. The fifty (50) S&A assemblies from 4.3.1 (c) shall be subjected to the test for presence of sliders and slider springs specified in 4.5.3. The assemblies shall then be subjected to the arming test specified in 4.5.1, followed by the non-arming test specified in 4.5.2.

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d. The eighty (80) sensor assemblies from 4.3.1 (d) shall be subjected to the tests specified in 4.5.7.1 through 4.5.7.5. An additional ten (10) sensor assemblies from 4.3.1 (d) shall be conditioned to -25 + 5°F and subjected to the release test specified in 4.5.7.2 for informational purposes. The pressure for this test shall be 300 + 25 psi applied for 1 second maximum.

e. The twenty-five (25) AD switches from 4.3.1 (e) shall be subjected to all the examinations and tests specified in 4.4.2.12.

f. The two hundred sixteen (216) housings from 4.3.1 (f) shall be subjected to the following:

1. All the tests specified in 4.5.12.

2. After completion of the tests specified in 4.5.12, the jumper wires shall be soldered in and the housings shall be subjected to the examinations of 4.4.2.23 after molding over the test pads.

g. After completion of the examinations in f above, the two hundred sixteen (216) housing shall be prepared as specified in 4.5.15. One test bar shall be cast from each molding compound batch used to bond the plug to the housing. The test bar(s) shall be subjected to the compressive strength test specified in 4.5.10.2. The assemblies shall be finished to comply with Dug. 9298399 inspected per 4.4.2.24. The 216 non-hazardous mines will be subjected to a ballistics acceptance test at a Government test facility.

4.3.3 Rejection. The first article quantity shall be rejected if any of the following occurs:

a. Any test bar fails to comply with the compressive strength requirement for the molding compound. NOTE: If a test bar fails due to a void, a second test bar from the same batch may be tested in its place.

b. Any barrier assembly fails any of the examinations specified in 4.4.2.2 or any barrier assembly fails the shear test specified in 4.5.5.

c. Any S&A assembly fails the arm or non-arm test or any S&A assembly fails all examinations specified in 4.4.2.4 excluding Major Defect 101.

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d. The sensor assembles subjected to the functioning tests exceed the reject criteria specified in the applicable paragraphs.

Any AD switch fails any of the examinations or tests specified in 4.4.2.12.

f. Any housing fails to comply with any of the electrical tests specified in 4.5.12 (no defects permitted) or the examinations in 4.4.2.23.

g. The ballistic sample exceeds the allowable defects (4.4 .3.2 - 4.4.3.7).

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Inspection lots of Housing, Timing and Fuzing shall contain:

Electric detonators (Dwg. 9245691) of one lot interfix number from one manufacturer (See 4.5.6).

b. Lead cup assemblies (Dwg. 9298456) of one lot interfix number from one manufacturer (See 4.5.6).

c. Battery assembly (MIL-B-48343) from one manufacturer.

d. Molding compound (MIL-M-48245) from one manufacturer.

e. Gas generator (Dwg. 9298530) of one lot interfix number from one manufacturer.

f. Integrated Circuits (Dwg. 9298507) from one manufacturer.

g. Thick Film Resistor Network (Dwg. 9298496) from one manufacturer .

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor’s quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Where cited herein, attributes sampling inspection shall be conducted in accordance with Table I below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE I Attributes sampling inspection.

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

Lot Size	Special Inspection Levels S-I	S-II	S-III	S-IV
2 to 15	*	*	*	13
16 to 50	*	*	32	13
51 to 150	*	125	32	13
151 to 500	*	125	32	13
501 to 3200	1250	125	32	13
3201 to 35000	1250	125	32	13
35001 to 500000	1250	125	32	13
500001 and over	1250	125	32	32

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by contractor when they provider as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein.

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In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Printed Wiring Board, Electronics		9298653
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Color incorrect for time (Natural board with long time path)	100%	3.2	Visual
Major				
101	Blue board with short time path	100%	3.2	Visual
102	Printed wiring board material		3.1	Certification
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Barrier Assembly		9298476
			NEXT HIGHER ASSEMBLY 9298464

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness minimum (min.)	Level III	3.2	Gage
102	Diameter, min.	Level III	3.2	Gage
103	Material (See Note 1)		3.1	Certification
104	Shear test	20-0-1	3.2	4.5.5
105	Presence of burrs or sharp edges	Level III	3.2	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:. Material certification required for each lot of barrier assemblies.

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	S&A Assembly	1 OF 1	9298464
			NEXT HIGHER ASSEMBLY 9298453

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Non-arming test	100%	3.2	4.5.2
Major				
101	Arming test	100%	3.2	4.5.1
102	Long or short slider missing or improperly positioned	100%	3.3.1	4.5.3
103	Long or short slider spring missing or improperly positioned	100%	3.3.1	4.5.3
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	S&A Assembly (After completion of arming and non-arming tests)	1 OF 1	9298464
			NEXT HIGHER ASSEMBLY 9298453

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical 1	Both barrier assemblies missing or improperly positioned	100%	3.2	Gage (Note 1)
Special a	Either barrier assembly missing or improperly positioned	100%	3.2	Gage (Note 1)
Major 101	Setback pin of S&A Assembly present and operative	100%	3.2	4.5.4
Minor 201	Stakes missing or incomplete as evidence by mating holes in housing and side plate not completely covered (4 places)	Level V	3.2	Visual
202	Poor workmanship	Level V	3.20	Visual

NOTES: 1. All S&A Assemblies which have both barrier assemblies present and properly positioned (accepted by inspection for defects Critical 1 and Special a) shall be suitably marked such that the mark will remain visible after molding in the housing, timing and fuzing.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	S&A Assembly	1 OF 1	9298464
			NEXT HIGHER ASSEMBLY 9298453

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Barrier probe acceptance mark missing	100%	3.2	Visual
Major				
101	Any sealing ball or washer missing	200-2-3	3.2	Visual
102	Detonator and lead cup assembly qualification test	50-1-2	3.3.2	4.5.6, Note 1
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:
1. In the event of a qualification test failure a detonator serial lot may only be retested one time and this retest must be with a different lead cup serial lot.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.6	TITLE S&A Assembly and Laminate Assembly	SHEET 1 OF 1		DRAWING NUMBER 9298453
				NEXT HIGHER ASSEMBLY 9298451
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Lead cup assembly above flush	Level III	3.2	Gage
102	Lead cup assembly inverted	200-2-3	3.2	Visual
103	Sealant or foreign material on exposed surface of lead cup assembly	Level III	3.2	Visual
104	Shorting washer present (See Note 1)	Level III	3.2	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES: 1/ Flex circuit wired to S&A at the preform assembly level.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Manifold Assembly, Center	1 OF 1	9298547
			NEXT HIGHER ASSEMBLY: 12927051

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any passage blocked	Level III	3.2	Test
102	Tubing punctured, cut, loose or improperly assembled	Level III	3.2	Visual
103	Ball missing or loose	Level III	3.2	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-48189D(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Manifold Assembly, Side		9298543
			NEXT HIGHER ASSEMBLY 12927051

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any passage blocked	Level III	3.2	Test
102	Tubing punctured, cut, loose or improperly assembled	Level III	3.2	Visual
103	Ball missing or loose	Level III	3.2	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Diaphragm		9298461
			NEXT HIGHER ASSEMBLY 9298596

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of tears or cracks after mandral test	500-5-6	3.2	Visual
102	Material	Note 1		Certification
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:
1. Certification of compliance for material shall include the 100% virgin material requirement.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Sensor Assembly	1 OF 1	9298569
			NEXT HIGHER ASSEMBLY 12927051

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	Functioning Tests (Note 1)			
101	Non Release (Note 2)	315-3-4	3.4.1	4.5.7.1
102	Release and thread security	80-7-8	3.4.2	4.5.7.2
103	Tripline deployment	80-3-4	3.4.3	4.5.7.3
104	Breakwire functioning	80-5-6	3.4.4	4.5.7.4
105	Post retention and thread breaking strength	80-5-6	3.4.5	4.5.7.5
106	Soldered Connections Improper	Level III	3.8.1	4.5.7.6
107	Post locking ball missing (Note 2)	315-21-22	3.2	X-Ray
108	Sensor Intrusion Test	80-4-5	3.4.5	4.4.4/4.5.7.7
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTE 1: The tests shall be performed in the order listed. The sample units for each test shall have been tested on all preceding tests in the sequence. However, acceptance or rejection on one test shall not constitute acceptance or rejection on the other tests. The sample sizes specified are applicable to each test and shall not be amended by fallout from previous tests.

NOTE 2: This test to be performed prior to assembly of the breakwire, breakwire retaining ring, switch post cover and epoxy. The testing may be conducted as a normal production operation with an inspector verifying the operation.

NOTES.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Can, Switch	1 OF 1	9298515
			NEXT HIGHER ASSEMBLY 9298514

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Plating	5 per batch	3.2	4.5.8.3
Minor 201	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Switch Assembly	1 OF 1	9298512
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Switch function seven (7) contact closures minimum each direction	100%	3.2	4.5.8.1
102	Contact closure duration 0.60 seconds maximum	100%	3.2	4.5.8.1
103	Insulation resistance	100%	3.2	4.5.8.2
104	Assembly damaged, distorted or incomplete	Level III	3.2	Visual
105	Ribbon improperly welded to can	Level III	3.2	Visual
106	Epoxy seal missing or incomplete	Level III	3.2	Visual
107	Solderability, 2 leads	Level III	3.8	Visual (Note 1)
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES: Verify double dip operation each shift.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Battery and Shorting Bar Support Assembly	1 OF 1	9298669
			NEXT HIGHER ASSEMBLY 12927051

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Battery missing	100%	3.2	Visual
102	Leads not welded to battery	100%	3.2	Visual
103	Shorting bar missing	100%	3.2	Visual
104	Weld shear test		3.2	4.5.9
Minor	None defined			

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.14	TITLE Transistor, Detonator, NPN	SHEET 1 OF 1	DRAWING NUMBER 9298555
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Forward-current transfer ratio, HFE @ 85°C (Note 1)	100%	3.2	MIL-STD-750, Method 3076
102	Collector-emitter cutoff current, (Note 1)	100%	3.2	MIL-STD-750, Method 3041
103	Base-emitter voltage VBEF @ 85°C (Note 1)	100%	3.2	MIL-STD-750, Method 3066
104	Collector emitter saturation voltage, VCES @ 85°C (Note 1)	100%	3.2	MIL-STD-750, Method 3071
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:
NOTE 1: Tests to be performed at 85°C after subjecting the components to twenty-five (25) cycles of temperature shock per MIL-STD-883, Method 1010, Condition C.
NOTE 2: Only those transistors successfully completing the above tests shall be used in 9298495 electronics assembly.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.15	TITLE Capacitor, Tantalum, Dual, Dip Case	SHEET 1 OF 1	DRAWING NUMBER 9298503
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	Thermal Shock (Note 1)			
101	Capacitance	Level III	3.2	Note 4B/Dwg. 9298503
102	Dissipation	Level III	3.2	Note 2A/Dwg. 9298503
103	DC leakage (ambient temperature)	500-5-6	3.2	Note 2C/Dwg. 9298503
104	DC leakage (plus 85°C)	500-5-6	3.2	Note 2D/Dwg. 9298503
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES: 1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with Note 4B of Dwg. 9298503.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.16	TITLE Capacitor, Fixed, Tantalum, Solid, Polarized	SHEET 1 OF 1	DRAWING NUMBER 9298509
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	Thermal Shock (Note 1)			Method 107, Cond. A MIL-STD-202
101	Capacitance	500-5-6	3.2	Note 3B1/Dwg. 9298509 Method 305, MIL-STD-202
102	Dissipation factor	500-5-6	3.2	Note 3B2/Dwg. 9298509 Method 305, MIL-STD-202
103	DC leakage at 6VDC, (4 us Max.)	500-5-6	3.2	Note 3B3/Dwg. 9298509
Minor 201	Poor workmanship	Level V	3.20	Visual

NOTES: 1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with Method 107, Test Condition A of MIL-STD-202.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Capacitor, Fixed Plastic Dielectric	1 OF 1	9298658
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	Thermal Shock (Note 1)			Note 4B/Dwg. 9298658
101	Capacitance	500-5-6	3.2	Note 2A/Dwg. 9298658
102	Dissipation factor	500-5-6	3.2	Note 2C/Dwg. 9298658
103	Insulation resistance (ambient temperature)	500-5-6	3.2	Note 2D/Dwg. 9298658
104	Insulation resistance (plus 85°C)	500-5-6	3.2	Note 2D/Dwg. 9298503
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:
1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with Note 4B of Dwg. 9298658.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Resistor, Network, Thick Film	1 OF 1	9298496
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of MIL-R-63993		3.2	Certification
Minor 201	None defined			

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Integrated Circuit, Control and Timing Base CMOS	1 OF 1	MIL-I-48331/9298507
			NEXT HIGHER ASSEMBLY 9298495

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of MIL-I-48331		3.2	Certification
Minor 201	None defined			

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2C	TITLE Printed Wiring Board Assembly	SHEET 1 OF 1	DRAWING NUMBER 9298495
			NEXT HIGHER ASSEMBLY 9298490

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Solder workmanship	Level III	3.2	Visual
102	Land paths broken, raised or bridged with solder	Level III	3.2	Visual
103	Printed wiring board cracked or broken	Level III	3.2	Visual
104	Components damaged or missing	Level III	3.2	Visual
105	Components mounted with wrong polarity	Level III	3.2	Visual
106	Interconnect pins (8) present	Level III	3.2	Visual
Minor				
201	Maximum height	Level V	3.2	Gage
202	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Preform Assembly		9298451
			NEXT HIGHER ASSEMBLY 9298450

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Soldered connections improper or unsoldered	Level III	3.8	Visual
102	Retainer and ball assembly not seated on battery	Level III	3.2	Visual
103	Any component or assembly not secured to preform	Level III	3.2	Visual
104	Kinked, broken, loose or flattened sensor tubing	Level III	3.2	Visual
105	Any unconnected sensor tubing	Level III	3.2	Visual
106	S&A shorting washer removed	Level III	3.2	Visual
107	Presence of shorting buss with terminal pad leg intact	Level III	3.2	Visual (Note 1)
108	Gas generator assembly missing	Level III	3.2	Visual
109	Shorting bar plug missing	Level III	3.2	Visual
110	Shorting buss not soldered to pads	Level III	3.2	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:
1. For safety purposes, The S&A shorting washer must be present on the lead of the M100 detonator and shorted to the clip lead until just prior to soldering flex circuit to detonator leads and conducting detonator resistance test (see 4.5.12).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.22	TITLE Housing, Timing and Fuzing (Prior to molding over conductor test pads)	SHEET 1 OF 1	DRAWING NUMBER 9298450
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Type II housing (Blue Board) with hole in shorting bar (Blue Board must have bar without hole)	100%	3.2	Visual or approved Gage
Special				
a	Type I housing (Natural Board) without hole in shorting bar (Natural Board must have shorting bar with hole)	100%	3.2	Visual or approved Gages
Major				
101	Temperature control during cure and cure time (Note 1)	See Note 1	3.7	4.5.11
102	Resin/Hardener weight ratio of molding compound		3.2	4.5.10.1
103	Compressive strength		3.16	4.5.10.2
104	Shelf life of molding compound expired	Per batch	3.2	Visual
105	Functional testing of electronic circuitry	100%	3.5	4.5.12
106	Open shorting buss terminal pad leg, with a min. separation of .025 before filling with molding compound	Level III		Gage
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES Cure temperature and cure time shall be monitored for each molding compound batch of housings.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.23	Housing, Timing and Fuzing	1	3 9298450
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Barrier probe acceptance mark on S&A missing	100%	3.2	Visual or approved Gages
b	Shorting bar broken or damaged	100%	3.2.1	Visual or approved Gages
c.	Overall height (Note 2)	100%	3.2	Gage
d.	Perpendicularity of end face to apex (Note 2)	100%	3.2	Gage
e.	Profile envelope measurement	100%	3.2	Gage
Major				
101	Height of bare battery ball above apex radius	Level III	3.2	Gage
102	Height of shorting bar above apex radius	Level III	3.2	Gage
103	Voids in excess of specified criteria	Level III	3.9	Visual
104	Detonator shield missing or loose	Level III	3.2	Visual
105	Any sensor locating plug missing (7 places)	Level III	3.2	Visual
106	Arc face to apex contour minimum	Level III	3.2	Gage

NOTES:
1. Cure temperature and cure time shall be monitored for each batch of housings.
2. Perform 100% production screen with approved automatic gaging machine.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Housing, Timing and Fuzing	2 OF 3	9298450
			NEXT HIGHER ASSEMBLY N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	cont'd			
110	Molding material or other foreign material on exposed surface of lead cup assembly over .005	Level III	3.20.5	Visual
111	Incomplete fill in test pad area	Level III	3.2	Visual
112	Temperature control during cure min cure time	Note 1	3.7	Note 1
113	Distance from end face to forward edge of shorting bar	Level III	3.2	Gage
114	Distance from base of largest cavity	Level III	3.2	Gage
115	Color incorrect	Level III	3.10	Visual
116	Curved apex surface between battery ball and shorting bar	Level III	3.2	Gage
117	Rust on HTF Assembly	Level III	3.20.4	Visual
Minor				
201	Plug under shorting bar missing	Level V	3.2	Visual
202	Molding material in dovetail of S&A or booster slot	Level V	3.2	Visual
203	Loose end face laminates	Level V	3.2	Visual/Manual
204	Arc face to apex contour, maximum	Level V	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET OF 3	DRAWING NUMBER
4.4.2.23	Housing, Timing and Fuzing		9298450
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor	cont'd			
205	Any exposed tubing which is twisted, kinked, punctured or broken	Level V	3.2	Visual
206	Surface preparation of largest diameter of cavity missing or incomplete	Level V	3.2	Visual
207	Any markings missing, incorrect or incomplete	Level V	3.2	Visual
208	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Munition for Ballistic Test	1 OF 1	9298399
			NEXT HIGHER ASSEMBLY 9298319

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Battery activation during molding/ cure	100%	3.2	Voltmeter
202	Height	100%	3.2	Gage
203	Arc face radius	100%	3.2	Gage
204	Perpendicularity of arc face to triangular face	100%	3.2	Gage
205	Voids in bond area	100%	3.2	Visual
206	Distance from arc face to apex gaging radius	100%	3.2	Gage
207	Resin/Hardner weight ratio of molding compound		3.2	4.5.10.1
208	Molding compound shelf life expired	Per batch	3.2	Visual/ certification
209	Temperature control during cure and cure time	Note 1	3.2	4.5.11/certification
210	Any evidence of battery activation	100%	3.2	Visual
211	Poor workmanship	100%	3.20	Visual

NOTES: Cure temperature and cure time shall be monitored for each batch of munitions.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Packaging and Packing Material	1 OF 1	9313636 & 9313637
			NEXT HIGHER ASSEMBLY: 9313638

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Presence of foreign material	Level III	3.20	Visual
102	Material damaged (broken, torn, punctured, etc.)	Level III	3.20	Visual
103	Evidence of water damage or excessive moisture	Level III	3.20	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Packaging and Packing Housings	1 OF 1	9313636 & 9313637
			NEXT HIGHER ASSEMBLY: 9313638

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Type II housing(s) present in Type I housing lot	100%	3.20	Visual
Major				
101	Type I housing(s) present in Type II housing lot	100%	3.20	Visual
102	Housing(s) incorrectly positioned	100%	3.20	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Packaging and Marking for Box, Fiberboard for Housing, Timing and Fuzing, Type I/Type II	1 OF 1	9313638
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Exterior markings indicate Type I housings, but contents are Type II housings	100%	3.20	Visual
Major				
101	Exterior markings indicate Type II housings, but contents are Type I housings	Level III	3.20	Visual
102	Box damaged	Level III	3.20	Visual
103	Markings missing, illegible, incorrect, or incomplete	Level III	3.20	Visual
Minor				
201	Poor workmanship	Level V	3.20	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Special Test Mine, Type I & II	1 OF 1	9298399
			NEXT HIGHER ASSEMBLY 9298399

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Loading operation	Note 1	3.6	4.4.3.2
Minor 201	Poor workmanship	Level V	3.20	Visual

NOTES:
1. Verify loading operation is accomplished per the specified requirements.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.29	Special Test Projectile for M692/M731 155MM	1 OF 1	9298319
			NEXT HIGHER ASSEMBLY 9298319

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Loading operation	Note 1	3.6	MIL-T-70638
Minor 201	Poor workmanship	Level V	3.20	Visual

NOTES: 1. Verify loading operation is accomplished per the specified requirements.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Special Test Projectiles	OF 1	9298319
			NEXT HIGHER ASSEMBLY 9298319

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Projectile temperature		3.14	4.4.3.1.1/Visual
102	Projectile firing zone and temperature		3.15	4.4.3.1.1/Visual
Minor	None defined			

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.31	TITLE Special Test Mine Type I or Type II	SHEET 1 OF 1	DRAWING NUMBER 9298399
			NEXT HIGHER ASSEMBLY 9298399

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Hazardous dud	100%	3.13.2	4.5.19
Major				
101	Anti-disturbance functioning		3.11.1	4.4.2.3.2/4.4.3.5/4.5.17.1
102	Breakwire functioning		3.11.2	4.5.17.2
103	Tripline deployment		3.12	4.4.3.7/4.5.18
104	Safe duds		3.13.1	4.4.3.8/4.5.17.3
105	Early functions		3.11.4	4.4.3.9
106	Self destruct functioning		3.11.3	4.4.3.3/4.4.3.4/4.5.17.3
Minor	None defined			

NOTES.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	Wiring Assembly, Printed Flexible	1 OF 1	9298670
			NEXT HIGHER ASSEMBLY 12927051

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Electrical Test (continuity)	Level III	3.2	3.8.3.2
102	Electrical Test (insulation resistance)	Level III	3.2	3.8.3.3
103	Solderability (all lands)	10-0-1	3.2	3.4.6
Minor				
201	Diameter (all holes)		3.2	Gage
202	Poor workmanship	Level V	3.20	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3.1 Non-hazardous projectile firings. Six (6) projectiles per lot shall be fired for lot acceptance testing. The rounds shall be conditioned and fired as follows:

Temperature	Quantity
+70°F	2
-25°F	1
+70°F	2
+145°F	1

4.4.3.2 Projectile mine distribution. Each projectile mine cargo shall be divided so that one-third (1/3) of the recoverable mines are tested for anti-disturbance, one-third (1/3) of the recoverable mines are tested for breakwire, and one-third (1/3) of the recoverable mines are tested for self-destruct, Samples shall be preselected, numbered and color coded in accordance with 4.5.15.4.

4.4.3.3 Missing mines and accidental functioning. Any mine which has not been located by eighty (80%) percent of the maximum self-destruct time for the type of munition under test shall be classed as missing. Any mine which is functioned accidently by any means shall be classed as an accidental functioning. Both missing mines and accidental functioning shall be a no-test for all test requirements, except hazardous duds.

4.4.3.4 Self-destruct functioning. Mines selected for self-destruct functioning shall be tested for the specified criteria in accordance with 4.5.15.3. Scoring for self-destruct functioning shall be per 4.5.18. The lot shall be rejected if more than 7.2% of those recoverable units in the self-destruct sample set are classed as failure to function self-destruct.

4.4.3.5 Anti-disturbance functioning. Mines selected for anti-disturbance testing shall be tested for the specified criteria in accordance with 4.5.15.1. Scoring for anti-disturbance functioning shall be per 4.5.18. The lot shall be rejected if more than eleven percent 11.0% of those units tested for anti-disturbance functioning are classed defective. Safe duds shall be excluded from the anti-disturbance sample set.

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4.4.3.6 Breakwire functioning. Mines selected for breakwire functioning tests shall be tested for the specified criteria in accordance with 4.5.15.2. Scoring for breakwire functioning shall be scored in accordance with 4.5.18. The lot shall be rejected if more than 7.9% of those recoverable units in the breakwire sample set are classed defective as failing to function breakwire. Safe duds shall be excluded from the breakwire sample set.

4.4.3.7 Tripline deployment. All recoverable mines shall be examined for the specified criteria in accordance with 4.5.16. The lot shall be rejected if more than 4.2% of all recoverable mines are classed defective for tripline deployment. Safe duds shall be excluded from the tripline deployment sample set.

4.4.3.8 Safe duds. Any recoverable mine which fails to function prior to the maximum self-destruct time for the type of munition under test and which fails to function when the hazardous dud inputs specified in 4.5.17 are applied shall be classed a safe dud. The lot shall be rejected if more than 4.0% of all recoverable mines are classed defective as safe duds.

4.4.3.9 Early function. Any recoverable mine which functions self-destruct prior to the minimum self-destruct time period for the type of munition under test shall be classed as an early function. The lot shall be rejected if more than 4.9% of all recoverable mines, which are designated as M731, are classed defective for early function. If the mines being tested are designated as M692 mines, the lot shall be rejected if more than 5.8% of all recoverable mines are classed defective for early function.

4.4.4 Sensor intrusion test. Each lot of sensors shall be tested for the epoxy intrusion requirement (defined by 3.4.6 and 4.5.7.7) to determine if there is any epoxy intrusion. Testing for this requirement shall be completed prior to lot acceptance testing for sensor functional requirements (as defined in 3.4.2 thru 3.4.5). The sampling plan used for the intrusion test defined in 3.4.6 shall be 80-4-5. Any lot failing to comply with the intrusion sampling shall be rejected.

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4.4.4.1 Sensor wire continuity. As an alternate to inspecting a sample of the parts, the contractor may perform an inspection to verify the effectiveness of a 100% production screen for continuity of the sensor circuits. The inspection must verify that the test equipment and production operator are making the correct decision during the 100% screening operation. The plan for this verification must be documented and forwarded to the Technical Agency for approval (see 6.4).

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the tables beginning with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection.

4.5.1 Arming of S&A assembly. The S&A assembly, with a barrier restriction pin in place, shall be mounted in the arming nest of the arming fixture and the arming conditions specified shall be applied. observation for arming shall be made as evidenced by both Barrier Assemblies being in the out-of-line position. Each unit so tested must be recocked by placing the assembly in the recock nest of the arming fixture and subjecting the assembly to the specified recocking conditions. Each unit failing to arm shall be removed from the lot.

4.5.2 Non-arming of S&A assembly. The S&A assembly, shall be mounted in the arming nest of fixture and subjected to the non-arming conditions as specified. observation for non-arming shall be made as evidenced by both Barrier Assemblies being in the fully unarmed position. Each unit failing this test shall be removed from the lot.

4.5.3 Presence of sliders and slider springs in the S&A assembly. The S&A assembly shall be radiographer with the axis of radiation in line with the axis of the lead cup and detonator holes. Units which fail to meet the requirement of 3.3.1 shall be classed defective and removed from the lot. This test is non-destructive . An alternative inspection method may be used provided that it can determine presence and position of sliders and slider springs. Any alternative method must be submitted as specified in 4.4.5. Radiographic equipment operations and procedures shall be qualified in accordance with MIL-STD-453 excluding the “marking” requirement. In addition, the sensitivity of radiographic techniques shall be at least two (2) percent, and the photographic density of the film used shall be of a quality which will produce a sharp definition of internal structure.

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4.5.4 Setback pin of S&A assembly, present and operative. The S&A assembly, less detonator lead cup assembly, sealing balls and sealing washers, shall be tested using a fixture(s) which incorporate the forces as specified in drawing 9298464. The test shall be conducted in a sequence where the lesser force is applied last. Any unit failing to meet either force requirement shall be classified defective, suitably identified, and removed from the lot.

4.5.5 Shear test of the barrier assembly. The barrier assembly shall be placed in an approved fixture and the load specified on Drawing 9298476 shall be applied. any barrier assembly that completely separates shall be classed defective. This test is destructive . Units so tested shall not be returned to the lot.

4.5.6 Qualification test of electric detonator and lead cup assembly lots. This qualification test shall be performed for that combination of detonator serial lot and lead cup assembly serial lot components to be used for this test shall be randomly selected from the respective lots. All detonators selected for this test shall remain shorted, (leads twisted) unit immediately prior to firing. The components shall be assembled into S&A's comprised of housing, center plate, gasket and detonator side plate. (Sliders, springs, barrier assemblies, and setback pin and spring are not required). The S&A's shall meet applicable drawing requirements, including those affecting the gap between the detonator and lead cup assembly. Detonating cord assemblies from an accepted lot shall be suitably attached next to the lead cup assembly with a .020 inch minimum spacing between the two assemblies. The short shall then be removed and the detonators shall be initiated from behind an approved barricade using a suitable power source (a 100 microfarad capacitor charged to 1.6 volts). Observation shall be made for failure to initiate the detonating cord assembly. Approved barricades shall be used for assembling and testing these components . If the number of assemblies failing to initiate the detonating cord exceeds the specified acceptance criteria, that lot combination of detonators and lead cup assemblies shall be rejected for use in the housing, timing and fuzing.

4.5.7 Sensor assembly.

4.5.7.1 Non-release of sensor assembly. The sensor assembly less breakwire, retaining ring cover, and sealant, shall be placed in a centrifuge and spun in an orientation to apply the force specified in 3.4.1 along the direction of deployment. Any unit in which the sleeve completely separates or begins to separate from the case shall be classified defective and removed from the lot. This test is non-destructive. Those units successfully completing this test shall be used to make complete sensor assemblies.

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4.5.7.2 Release and thread security of sensor assembly. The breakwire resistance shall be checked for continuity. Units which have resistance greater than 1,000 ohms shall not be subjected to this test. The units to be tested shall be placed in a suitable fixture and the pressure specified in 3.4.2 shall be applied. All units shall be classified defective if any of the following occur:

a. Fails to release when initially pressurized and evidenced by the bobbin core being retained within the sensor case.

b. The breakwire continuity is broken as evidenced by a resistance greater than 1,000 ohms across the exposed ends of the breakwire.

c. The thread does not remain attached to the bobbin and post.

d. The thread breaks during release.

e. Any unit that does not release when initially pressurized, but releases when the pressure is reapplied, shall be classed defective. This test is destructive. Those units capable of meeting the requirements of the test specified in 4.5.7.3 shall be used for that test even if they failed to meet the requirements for this test.

4.5.7.3 Tripline deployment of sensor assembly. After the sensor assembly has been tested in accordance with 4.5.7.2, and with the sensor remaining in the test fixture, observation shall be made to assure that the bobbin core, with tripline attached, is deployed over a vertical barrier oriented as follows:

a. The barrier location shall be 16 feet, minimum, measured horizontally from the sensor.

B. Barrier height shall be 7 inches, minimum, above the top of the sensor. A bobbin core that impacts ground level before the barrier or impacts the barrier shall be considered defective. This test is destructive.

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4.5.7.4 Breakwire function of sensor assembly. After the sensor assembly is tested in accordance with 4.5.7.1, 4.5.7.2, 4.5.7.3 and with the unit remaining in the test fixture, a suitable force gage shall be attached to the tripline at a point just above the barrier. A pull force shall be applied along the axis of the bobbin post and the breaking force shall be noted. The breaking force is that force at which the resistance across the exposed ends exceeds the limit specified in 4.5.7.2. Any sensor assembly which does not meet the requirements specified in 3.4.4 shall be classed defective. This test is destructive. Those units that are capable of meeting the requirements of the test specified in 4.5.7.5 shall be used for that test even if they failed to meet the requirements of this test. The rate of pull shall be a person walking at a normal rate.

4.5.7.5 Post retention and thread breaking strength. After the sensor assembly is tested in accordance with 4.5.1.1, 4.5.7.2, 4.5.7.3, and 4.5.7.4, and with the unit remaining in-the test-fixture, the specified force shall be applied at a rate of 4 inches/see minimum to the tripline along the axis of the bobbin post. Any unit that releases the bobbin post or on which the tripline breaks shall be classed defective. This test is destructive. Units so tested shall not be returned to the lot. The rate of pull shall be a person walking at a normal rate.

4.5.7.6 Soldered electrical connections of sensor assembly. The electrical connections of the breakwire to the breakwire terminals shall be tested for electrical continuity using a suitable ohmmeter . The soldered connection shall have a smooth, bright appearance with metallic luster and shall not have a chalky, gritty surface, nor exhibit point (peaks), pits, scars, and fractures. The connections shall be completely covered by solder to the extent that no base metal is visible, with the exception that exposed copper is permissible on the end of the cut wires.

4.5.7.7 Molding material intrusion test. The sensor assembly will be placed into a test brick mold with (12) sensors per mold. Shrink tubing will be assembled to the sensor nozzles and exit the mold . Sensor plugs will be assembled to the cap end. The mold will be closed and subjected to the normal wedge molding process. The test brick cure will be for a minimum of 16 hours at 155° + 10°F cure temperature. The sensor caps shall be removed and the test brick assembled to an approved test fixture. The pressure specified in 3.4.6 shall be applied to the protruding end of the shrink tubing. This is a destructive test. Units may not be returned to the lot.

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4.5.8 Anti-disturbance switch.

4.5.8.1 AD switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 360° on its cylindrical axis, while this axis is held at 15 ± 2° referenced to the horizontal, for both CW and CCW directions of rotation. The functional requirements of drawing 9298512 shall be met.

4.5.8.2 Insulation resistance of AD switch. The switch assembly, when mounted in a suitable fixture, shall be measured for insulation resistance in the rest position using an approved megohmmeter. The megohmmeter reading shall meet the requirements of drawing 9298512. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.8.3 Plating of AD switch components. Five (S) components shall be randomly selected from each plating batch. The specimens shall be crossed-sectioned through the axial center of the sample, polished and examined microscopically to measure the plating thickness. If any sample fails to comply with the plating requirements, the plated batch shall be classed defective. (Destructive test)

4.5.9 Weld test of battery and shorting bar support assembly. A sample of one 1) part from the first ten (10) welded shall be tested to qualify the welding equipment for use in production. In addition, one (1) sample shall be tested after production of each one hundred (100) welded parts as a control of welding equipment during production. The units shall be placed in a fixture capable of applying the force specified on the applicable drawing. Any assembly in which the weld cracks or breaks when the specified force is applied shall be classed defective. If any defective parts are found during qualification of welding equipment, the equipment producing the defective assembly shall not be used in production. If any defective assembly is found when inspection is performed for control of the welding equipment, the equipment producing the defective weld shall be removed from production. Further, that portion of production since the last weld control check shall be returned to the contractor for corrective action. If welding is stopped for one (1) hour or longer the equipment shall be subjected to the qualification test prior to use in production. All welding equipment removed from production because of some fault may, after reworking, be returned to production, providing it passes the qualification test above. The contractor may use coupons of the same material and thickness as the parts being welded subject to approval of the Contracting Officer.

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4.5.10 Quality controls for each molding compound batch for housing.

4.5.10.1 Resin/hardener weight ratio control. At the beginning of production and at least once each four (4) hours of production, or portion thereof, the inspector shall draw two (2) test shots with the mixer set to expel the hardener and resin separately. The hardener and resin of each shot shall be weighed individually to verify that the ratio is between 1.8/1 and 2.2/1. If either shot fails to meet the requirement, all items filled since the last acceptable check shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

4.5.10.2 Mixing and cure control. One (1) test bar from each batch of molding compound (see 4 5.10.1) shall be molded at the same time the batch is used to mold housings, for lot acceptance testing. Vibration and evacuation as necessary may be used to mold the bar. The mix ratio limits shall be 2.0/1.0 ± 5% (1.8/1.0 to 2.2/1.0). The test bar shall be processed with the items molded from the batch of molding compound which the bar represents i.e., the bar shall be cured in the same ovens and subjected to the same temperatures for the same time periods as the items is represents. After completion of the cure, the bar shall be tested for ultimate compressive strength, as specified in ASTM D-695 (69) except that the specimen dimensions shall be 0.5 inch by 0.5 inch by 1.0 inch or 0.5 inch diameter by 1.0 inch long. The ends shall be flat and parallel to within .003 inch, and shall be perpendicular to the long axis of the specimen. Speed of testing shall be as specified in the ASTM. IF the results of this test are outside the limits specified in 3.16, the items represented by the bar shall be rejected.

4.5.11 Temperature control during molding/cure operations. Whenever the preform assembly, the housing, or munition is subjected to heating in-ovens, temperature records shall be maintained and reviewed to assure that the minimum temperature has been reached, and that the maximum temperature has not been exceeded. If any units have been subjected to temperatures beyond the maximum limit at any time, regardless of duration, they shall be rejected. Temperature-measuring instruments shall be approved and calibrated in accordance with 4.4.5.

4.5.12 Functioning of the electronic circuitry of the housing, timing and fuzing. For test procedures related to the functioning of the electronic circuitry of the housing, timing and fuzing. (See Appendix A of this specification.)

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Test	Paragraph	Limits
Detonator and Gas Generator Logic	40.2.4	Improper
Primary Oscillator Frequency	40.2.1	a) Less than 97% of nominal b) More than 104% of nominal c) Equal to or greater than 97% and less than 99% of nominal or greater than or equal to 104% of nominal
Barrier Oscillator Frequency	40.2.1	a) Less than 97% of nominal b) More than 104% of nominal c) Equal to or greater than 97% and less than 99% of nominal or greater than or equal to 102% but less than or equal to 104% or nominal
System Operating Current	40.2.2	a) Greater than 120 microamps b) Greater than 80 microamps but less than 120 microamps
Self-Destruct Time-Out	40.2.3	a) Greater than 100.1% of nominal b) More than 99.9% or nominal

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Test	paragraph	Limits
Low Voltage	40.2.5 At TA = +25 C ± 5.1 c (+77 ± 10 F)	a) Type I: Nonfunctioning, or functioning less than 2.8 volts or greater than 3.4 volts. Type II: Nonfunctioning, or functioning less than 2.8 volts or greater than 3.2 volts. b) Type I: Equal to or greater than 3.2 volts but less than 3.4 volts. Type II: Equal to or greater than 3.0 volts but less than 3.2 volts.
Current	40.2.13	a) Less than 4.5 milliamps b) Greater than 5.75 milliamps but less than or equal to 6.0 milliamps
Final Current	40.2.4	Less than 4.5 milliamps
Detonator Resistance	40.2.9 (See note)	a) Less than 2.9 ohms or greater than 9.5 ohms Equal to or greater then 8.5 ohms but less then 9.5 ohms

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NOTE: This test shall be 100% performed on P/N 9298451 prior to molding, and precaution should be taken to assure that the solder bridge is present to prevent inadvertent functioning detonator during handling prior to test. Assurance shall be provided that the solder bridge is removed immediately prior to resistance test so as not to invalidate test results.

Test	Paragraph	Limits
Output Capacitor Voltage	40.2.6	Improper
Continuity	40.2.12	a) Both circuits open b) One circuit open
AD Switch Operation	40.2.15	

The following tests shall be performed after adding the permanent detonator short:

Detonator Short	40.2.11	Absence of Short
Resistance	40.2.10	a) Equal to or less than 1.5 ohms or equal to or greater than 8.0 ohms b) Greater than 4.0 ohms but less than 8.0 ohms

4.5.13 Pull test. The sample shall be fixtured and the ribbon lead shall be pulled in the specified direction to meet the specified minimum force. The sample shall not indicate failure of the weld during the test. Any sample failing to comply with all of the specified requirements shall be classed defective. (Destructive test)

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4.5.14 Non-hazardous mine loading. The mine shall be loaded by a Government approved loading plant as designated by the contracting officer. Mines shall be loaded in accordance with drawing number 9298399 and MIL-T-70638.

4.5.15 Mine functioning.

4.5.15.1 Anti-disturbance functioning. The sample selected for anti-disturbance functioning shall be tested in the following manner.

a. The sample under test shall be tested as follows:

(1) If the mine is resting on an end face, rotate it to an adjacent slant face or arc face.

(2) If the mine fails to function on the first movement, it shall be moved a second time to adjacent face. If the mine is currently resting on a slant face, it can be moved to either the other slant face or end face not previously encountered in the first movement. If the mine is resting on the arc face it shall be rotated to the end face not encountered during the first movement.

b. All data shall be recorded.

c. If the mine fails to function, it shall be tested for breakwire activation in accordance with 4.5.15.2a through 4.5.17.2e.

d. If the mine fails to function, another tripline from the adjacent breakwire circuit shall be randomly selected and tested in accordance with 4.5.15.2a through 4.5.15.2e.

e. If the mine fails to function self destruct it shall be tested in accordance with 4.5.17 for the hazardous dud requirements of 3.13.2 (Destructive test).

f. The data acquisition shall include as a minimum, functioning, face mine is resting on after election from projectile, and face(s) involved in function movements. All data acquisition shall refer back to mine number identification.

4.5.15.2 Breakwire functioning. The sample selected for breakwire functioning shall be tested in the following manner:

a. The sample to be tested shall be restricted from any movement.

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b. Any tripline from either breakwire circuit shall be selected for an activation attempt.

c. A digital readout force gage shall be attached to the lead and an axial force shall then be applied to the device under test.

d. The tripline with gage attached shall be pulled in a straight line along the sensor cavity axis until the unit functions. The force at which the mine functions shall be recorded or the force at which the tripline thread or other components fail.

e. If the mine functions after breakage of the thread or other components, the mine is considered to have successfully functioned on the breakwire mode. The mine shall be classed defective if it fails to function in accordance with any of the specified criteria. Record all data.

In the event the mine fails to fire on first breakwire attempt, the following procedure shall be followed:

f. The mine shall be tested for anti-disturbance functioning in accordance with the procedures of 4.5.15.1 and the requirements of 3.11.1.

If the mine fails to function in the anti-disturbance mode, another tripline from the adjacent breakwire circuit shall be tested as noted above for breakwire functioning for information purposes only.

a If the mine fails to function in the second breakwire attempt, it shall be left to time out for self destruct functioning.

h. If the mine fails to function self destruct it shall be tested in accordance with 4.5.17 for the hazardous dud requirements of 3.13.2 (Destruction test).

4.5.15.3 Self destruct function. The mines selected for self destruct functioning shall be flagged and monitored for detonation. times. All times of detonations, early or normal, shall be recorded (Destructive test).

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4 .5.15.4 Test mine identification. Test mines shall be serialized to permit indentification by lot, round, and position within the projectile.

4.5.16 Tripline deployment. All samples shall be visually examined and measured when required for the tripline deployment criteria of 3.12. Any mine failing to meet the specified criteria shall be classed defective (Non-destructive).

4.5.17 Hazardous dud evaluation procedures. Any sample which has failed to tire the detonator when subjected to any predesignated tests for anti-disturbance, breakwire or self destruct shall be evaluated as follows:

a. Maximum self destruct time shall elapse prior to additional test inputs.

b. one tripline shall be pulled from each breakwire circuit. If the mine fires in either attempt it shall be classed as a hazardous dud.

c. If the sample under test does not fire in the breakwire mode it shall be tested for anti-disturbance functioning by moving it twice to adjacent mine faces. If it fires in any attempt to fire anti-disturbance it shall be classed as a hazardous dud,

d. If the mine does not fire in the anti-disturbance mode, it shall be tested for self-destruct functioning by allowing to remain undisturbed for a time period equal to that of the self-destruct time. If it fires prior to reaching the maximum self destruct time it shall be classed as a hazardous dud.

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e. If the mine fails to fire in the self-destruct mode, it. shall be classed as a safe dud and returned for failure analysis to the HTF manufacturer.

f. After completion of the hazardous dud inputs listed in 4.5.17a - 4.5.17e, the mine shall be placed in a secure area and allowed to time out to 110% of maximum self destruct time for projectile type under test. The time starts when the mines are placed in a secured area.

4.5.18 Mine scoring criteria. Results of mines so tested in accordance with 4.5.15 shall be scored in accordance with Tables 4-1 to 4-3.

a. Units selected for breakwire functioning shall be scored in accordance with Table 4-1.

b. Units selected for anti-disturbance functioning shall be scored in accordance with Table 4-2.

c. Units selected for self-destruct functioning shall be scored in accordance with Table 4-3.

d. Sample sets for earlies, safe duds, and hazardous duds shall include all recoverable mines.

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Table 4-1

Scoring of units tested for breakwire functioning

TEST RESULTS		SCORING
1. Early-out (goes SD before minimum time)	-	Early-out (4.4.2.31 Major 105)
2. No function on first BW attempt, functions subsequently (AD or SD)	-	BW failure (4.4.2.31 Major 102)
3. No function in any mode prior to maximum SD time	-	Safe dud (4.4.2.31 Major 104)
4. Functions after maximum SD time	-	Hazardous dud (4.4.2.31 Critical 1)
5. Functions on first BW attempt	-	BW success

LEGEND

BW- Breakwire
AD-Anti-Disturbance
SD-Self-Destruct

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Table 4-2

Scoring of units tested for anti-disturbance function

	TEST RESULTS	SCORING
1.	Early-out (goes SD before minimum time)	- Early-out (4.4.2.31 Major 105)
2.	No function during AD test, functions subsequently (BW or SD)	- AD Failure (4.4.2.31 Major 102)
3.	No function in any mode prior to maximum SD time	- Safe dud (4.4.2.31 Major 104)
4.	Functions after maximum SD time	- Hazardous dud (4.4.2.31 Critical 1)
5.	Functions during AD test	- AD success

LEGEND

BW- Breakwire
AD-Anti-Disturbance
SD-Self-Destruct

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Table 4-3

Scoring of units tested for self-destruct functioning

TEST OUTCOME	SCORING
1. Early-out (goes SD before minimum time)	- Early-out (4.4.2.31 Major 105)
2. No function prior to maximum SD	- SD failure (4.4.2.31 Major 106)
	- Safe dud (4.4.2.31 Major 104)
3. Functions after maximum SD time	- Hazardous dud (4.4.2.31 Critical 1)
4. Goes SD within time limits	- SD success

LEGEND

SD-Self-Destruct

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5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging of the Housing, Timing and Fuzing shall be in accordance with Dwg. 9313637.

5.2 Packing. The Housing, Timing and Fuzing shall be packed in accordance with Dwgs. 9313636 and 9313638.

5.3 Marking. Marking of the box for the Housing, Timing and Fuzing shall be in accordance with Dwg. 9313636.

5.4 Shipping. When shipments of units from more than one lot are made at one time, each lot shall be kept separate, and the division between lots clearly indicated to-prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The Housing, Timing and Fuzing covered by this specification is intended for use in the munition for Projectile, 155MM, M692/M731.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in section 2.1.

6.3 Data cards. (See MIL-A-48078). Distribution of data cards shall include the following: Commander, US ARDEC, ATTN: SMCAR-QAT-M (D), Picatinny Arsenal, New Jersey 07806-5000.

6.4 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research Development and Engineering Center, ATTN: SMCAR-QAT-M (D), Dover, New Jersey 07801-5000. Request that letter of submittal state contractor, contract number, specification number , item nomenclature and classification of defects or test paragraph. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.5 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the procuring contracting officer

6.6 Data requirements. A data requirement should be included via DD FOrm 1423 in the contract to provide for reporting to the Government the results of inspection of the designated characteristics.

6.7 Molding compound batch definition. A batch of molding compound is herein defined as that quantity of compound mixed at one time using base resin and hardener of one lot number from one manufacturer to mold housings in one continuous unchanged process. When a continuous type mixing process is used, a batch is that quantity of molding compound mixed and dispensed in one continuous unchanged process during a one(1) hour period or portion thereof using base resin and hardener of one lot number from one manufacturer.

6.8 Projectile functioning. The following malfunctions of the special test projectile shall result in all test mines contained in the subject projectile being considered non-recoverable and shall not be cause for rejection of the lot. Only recoverable mines will be subject to the projectile mine distribution per 4.4.3.2 and subsequent mine functioning per 4.4.3.3, 4.4.3.4, 4.4.3.5, 4.4.3.6, 4.4.3.7 and 4.4.3.8.

6.8.1 Premature burst. A premature burst is considered to have occured if the projectile functioning in the weapon or the time of flight is less than 1.9 seconds.

6.6.2 Nose fuze malfunction. Failure of the nose fuze to function at the pre-determined time after firing, resulting in failure to expel the projectile cargo over the target area.

6.8.3 Expulsion altitude. Functioning of the expulsion charge below four hundred (400) meters above ground level over the target area or greater than seven hundred and sixty five (765) meters over the target area.

6.8.4 Functioning at all zones. Satisfactory function of the projectile, when fired at any zone, is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the nose fuze and expulsion charge at the pre-determined time after firing. This includes removal of all shorting bars, depression of all battery balls, and expulsion of the entire cargo.

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6.9 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament, Research, and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Subject term (keyword) listing.

Bath Soldering
Molding compound
Integrated circuit
Digital
Monolithic
Silicon
Thickfilm

6.11 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345- A607)

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MIL-P-48189D (AR)
APPENDIX A

ELECTRONIC ASSEMBLY, TEST PROCEDURE

10. SCOPE

10.1 Scope. This Appendix details the test procedure to be used to electrically test the complete ADAM HTF Assembly prior to its shipment. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance .

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks.

20.1.2 Other Government documents, drawings, and publications.

Government Drawings

9298452	System Schematic
9298450	Housing Timing and Fuzing
9298495	Electronic Assembly

30. SUBMISSION

30.1 Test requirements. All ADAM HTF's containing electronic assembly 9298495 shall be tested in accordance with this document prior to shipment or performance evaluation. Only HTF's successfully passing this test shall be used for delivery against contract.

30.2 Absolute limits.

	VOLTAGE	CURRENT (RUN)	TEMPERATURE
MAXIMUM	6.0V	160.0 MICRO A.	+125°C
MINIMUM	--	--	-65°C

30.3 Operating limits (hot & cold).

	VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC , FREQ.
MAXIMUM	4.5 V	80.0 MICRO AMPS	+60°C(+65°C)	NOMINAL +5%
MINIMUM	2.7 V	--	-40°C	NOMTNAL -5%

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30.4 Operating limits (ambient).

VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC. FREQ.
4.0 V	40 MICRO AMPS	+25°C	NOMINAL ± 1%

30.5 VB rise time and tolerance. The rise time on VB shall be greater than 1 millsecond and less the 500 millisecond to 96% of its final value. Tolerance on V_B is ±.05 VOLTS.

40. SYSTEM TESTS PROCEDURE

System Tests shall be performed in accordance with Table I in a manner which meets the intent of the referenced requirement paragraph.

40.1 Table 1.

40.2 Test procedure.

40.2.1 Oscillator frequency. The oscillator frequency shall be measured by connecting the S.U.T. (System Under Test) as shown in Figure 1. The counter shall have an input impedence of 108 ohms or greater (C=10pf max) and the measurement shall represent a 10 period average. Frequency of both F1 and F2 shall be measured, and be within the limits of Table I. Input to F1 shall have a minimum of 1,500 fast counts applied and shall stabilize for 500 milliseconds minimum before the measurement is made. Test sequence: 1. Close SW1 and allow 1,500 fast counts to enter. 2. Open SW1 and wait 500 milliseconds minimum. 3. Close SW2 to F1. 4. Measure 10 period average of F1. 5. Close SW2 to F2. 6. Measure 10 period average of F2.

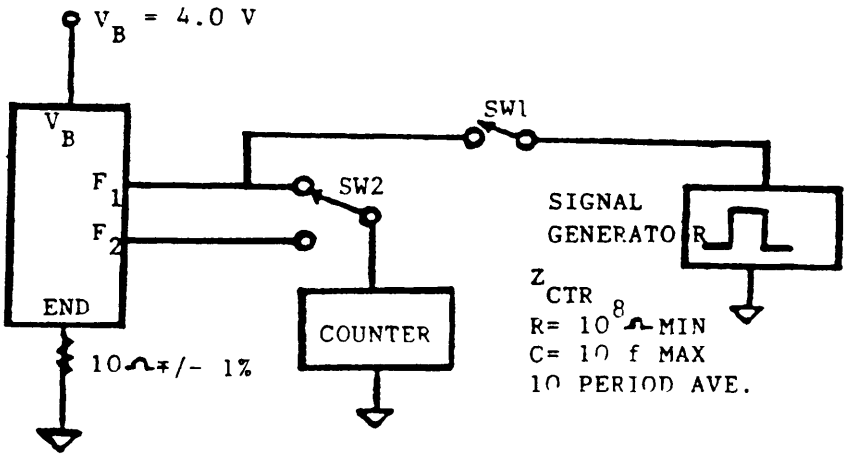


Figure 1. Oscillatory frequency measurement.

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APPENDIX A

Table I

GROUP I (TA=+25°C) QUALIFICATION AND LOT ACCEPTANCE TESTS

TEST	TEST CONDITION	REQUIREMENT PARAGRAPH	LIMITS MAX	MIN	UNITS
OUTPUT CAPACITOR VOLTAGE					
AFTER 1 SEC. CHARGE	V_B=3.3V	40.2.6.1	.200	.05	VOLTS
AFTER 14 SEC. DISCHARGE	V_B=3.3V	40.2.6.2		3.1	VOLTS
DETONATOR WAVEFORM	V_B=3.3V	40.2.4	Figure 6		
INITIATOR WAVEFORM	V_B=3.3V	40.2.4	Figure 6		
OSCILLATOR FREQUENCY	B_B=4.0V	40.2.1	+2%	-1%	*
SYSTEM OPERATING CURRENT	V_B=4.0V	40.2.2	80.0		MICRO AMPS
SELF DESTRUCT TIME OUT	V_B=3.3V	40.2.3	+1	-1	COUNT
L.V.D. LEVEL		40.2.5	3.2	2.8	VOLTS(Type 1)
			3.0	2.8	VOLTS(Type 2)
DETONATOR FIRING CIRCUIT	V_B=3.3	40.2.7	2.00		VOLTS(Turn on)
			1.00		VOLTS(Saturation)

*Period Value on drawing 9298452.

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APPENDIX A

Table I-Cont'd

GROUP I (TA=+25°C) QUALIFICATION AND LOT ACCEPTANCE TESTS

TEST	TEST CONDITION	REQUIREMENT PARAGRAPH	LIMITS MAX	MIN	UNITS
GAS GENERATOR FIRING CIRCUIT (PEAK)	V_B=3.3	40.2.8	2.00	VOLTS (TURN ON)	
			1.00	VOLTS (SATURATION)	
DETONATOR RESISTANCE		40.2.9	8.5	2.9	OHMS
GAS DETONATOR RESISTANCE		40.2.10	4.0	1.5	OHMS
DETONATOR SHORT		40.2.11	0.2		OHMS
BREAKWIRE CONTINUITY		40.2.12			
LOOP 1			HIGH	LOGIC LEVEL	
LOOP 2			HIGH	LOGIC LEVEL	
INITIAL CURRENT	V_B=4.0V	40.2.13	5.75	4.5	MA.
FINAL CURRENT	V_B=4.0V	40.2.14		4.5	MA.
A.D. SWITCH OPERATION		40.2.15	2.0		VOLTS

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Table II

GROUP II (TEMPERATURE-VOLTAGE LIMITS) QUALIFICATION TESTS ONLY

TEST	TEST CONDITION	REQUIREMENT PARAGRAPH	LIMITS MAX	LIMITS MIN	UNITS
TESTS TO BE RUN AT T_A=-40°C V_B=4.5V, 3.5V and T_A=60°C V_B=4.5V, 3.3V UNLESS OTHERWISE SPECIFIED.					
DETONATOR WAVEFORM		40.2.4		Figure 6	
GAS GENERATOR WAVEFORM		40.2.4		Figure 6	
OSCILLATOR FREQUENCY	V_B=4.2 UPPER LIMIT V_B=3.5 LOWER LIMIT	40.2.1	+5%	-5%	*
SYSTEM OPERATING CURRENT		40.2.2	80.0		MICRO A.
SELF DESTRUCT TIME OUT		40.2.3	+1	-1	COUNT
L.V.D LEVEL		40.2.5	3.5	2.7	VOLTS
DETONATOR FIRING CIRCUIT		40.2.7		1.6 .8	VOLTS(PEAK) VOLTS(1 MS)
GAS GENERATOR FIRING CURCUIT		40.2.8		1.6 .8	VOLTS(PEAK) VOLTS(! MS)
INITIAL CURRENT FINAL CURRENT	V_B=4.0V V_V=4.00V	40.2.13 40.2.14	5.75	4.0 4.0	MA MA

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40.2.2 System operating current. The system operating current shall measured by connecting the S.U.T. as shown in Figure 2. Input F_1 shall h 1500 fast counts applied with SW1 closed. Open SW1 and wait 100 millisec minimum and measure operative current. Operating current must meet the requirements of Table II.

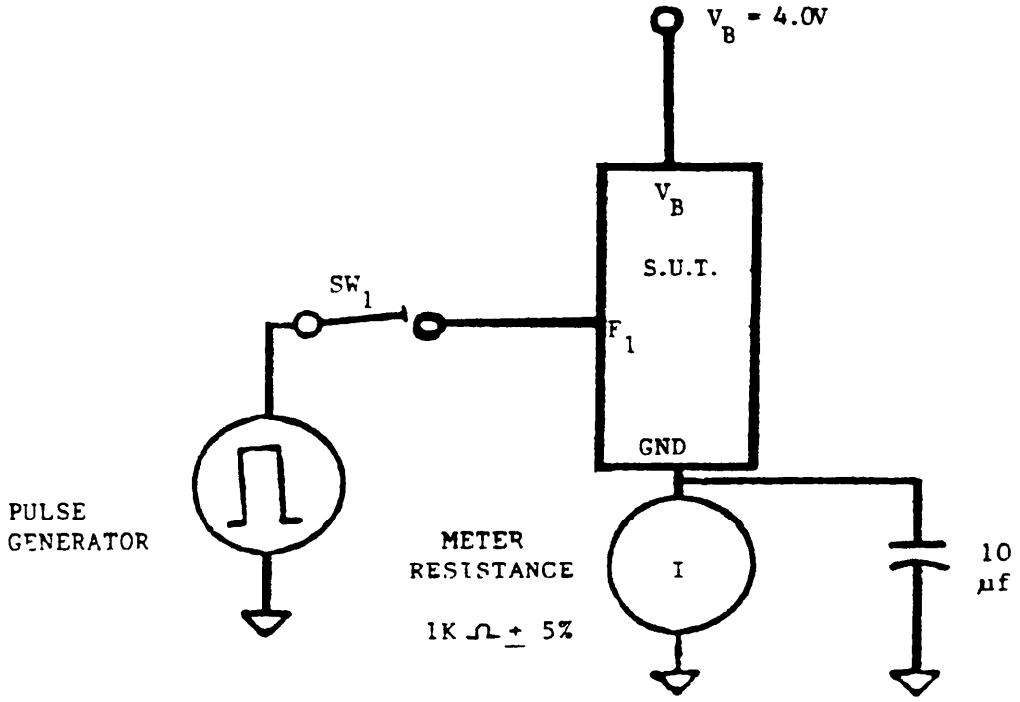


FIGURE 2. Operating current measurement.

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40.2.3 Self destruct time out. The self destruct time out shall be tested by connecting the S.U.T. as shown in Figure 3. Input F_1 shall have pulses applied at a rate consistent with the overall testing time desired and the output shall be observed at F.

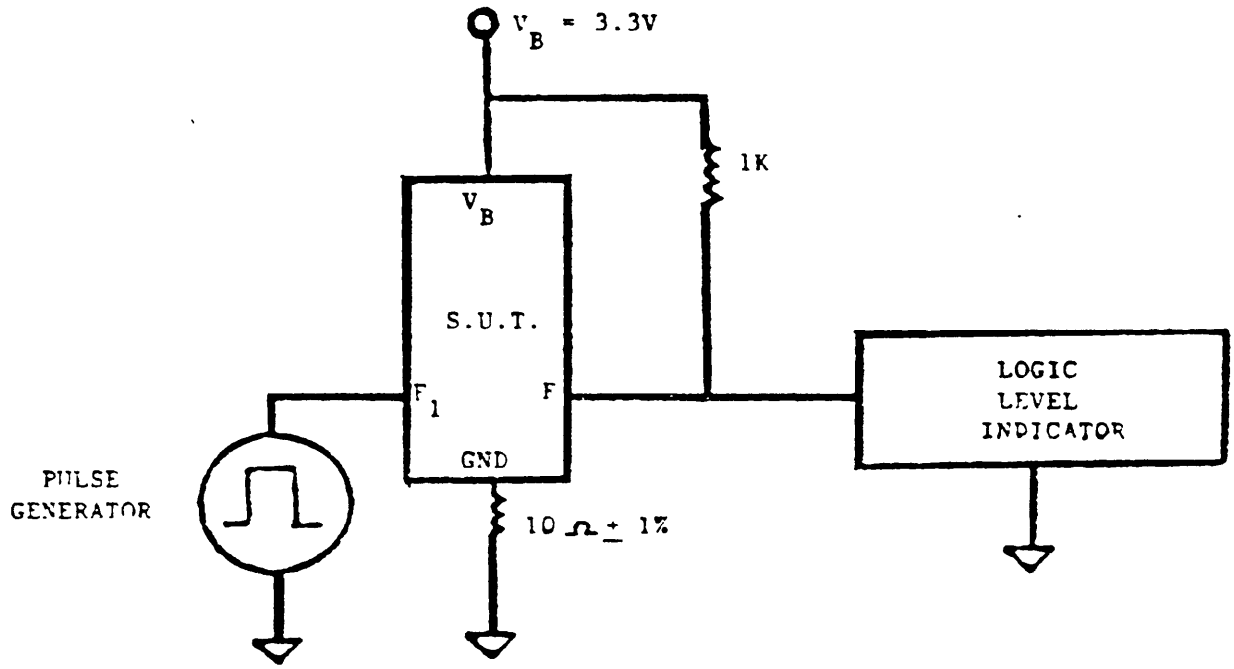


FIGURE 3. Self destruct time out measurement.

TEST SEQUENCE:

1. Connect S.U.T. as shown in Figure 3.
2. V_B turn on time shall be as specified in 30.5.
3. Pulse generator shall start when V_B=3.3V at a convenient repetition rate. The pulse generator shall have a duty cycle of 30% to 70%.
4. Output shall switch from a logic high (VB) to a logic low (GND) after the following input counts:

   A. Short timeout - 32000 counts
   B. Long timeout - 392448 counts

NOTE: If time out is measured by tiem rather than counts, then the limit shall be ± 0.1%.

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40.2.4 Detonator and gas generator logic. The S.U.T. shall be connected as shown in Figure 4. When the input waveforms shown in Figure 5 are applied, the output shown in Figure 6 shall occur.

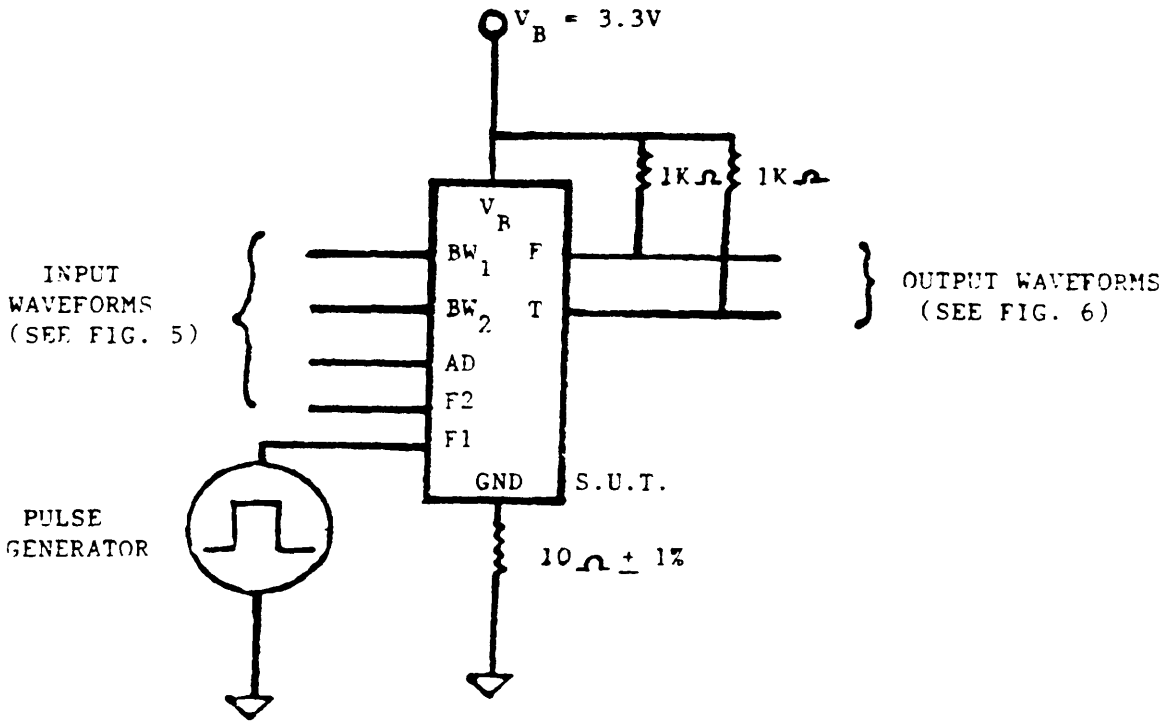


FIGURE 4. Detonator and gas generator output measurement,

The test sequence shown in Figure 5 and 6 may be changed provided all tests specified are accomplished. The order of accomplishment is not important. The "T" output waveform shall be present when F1 = 160 pulses. However, only one pulse shall be produced at “T”.

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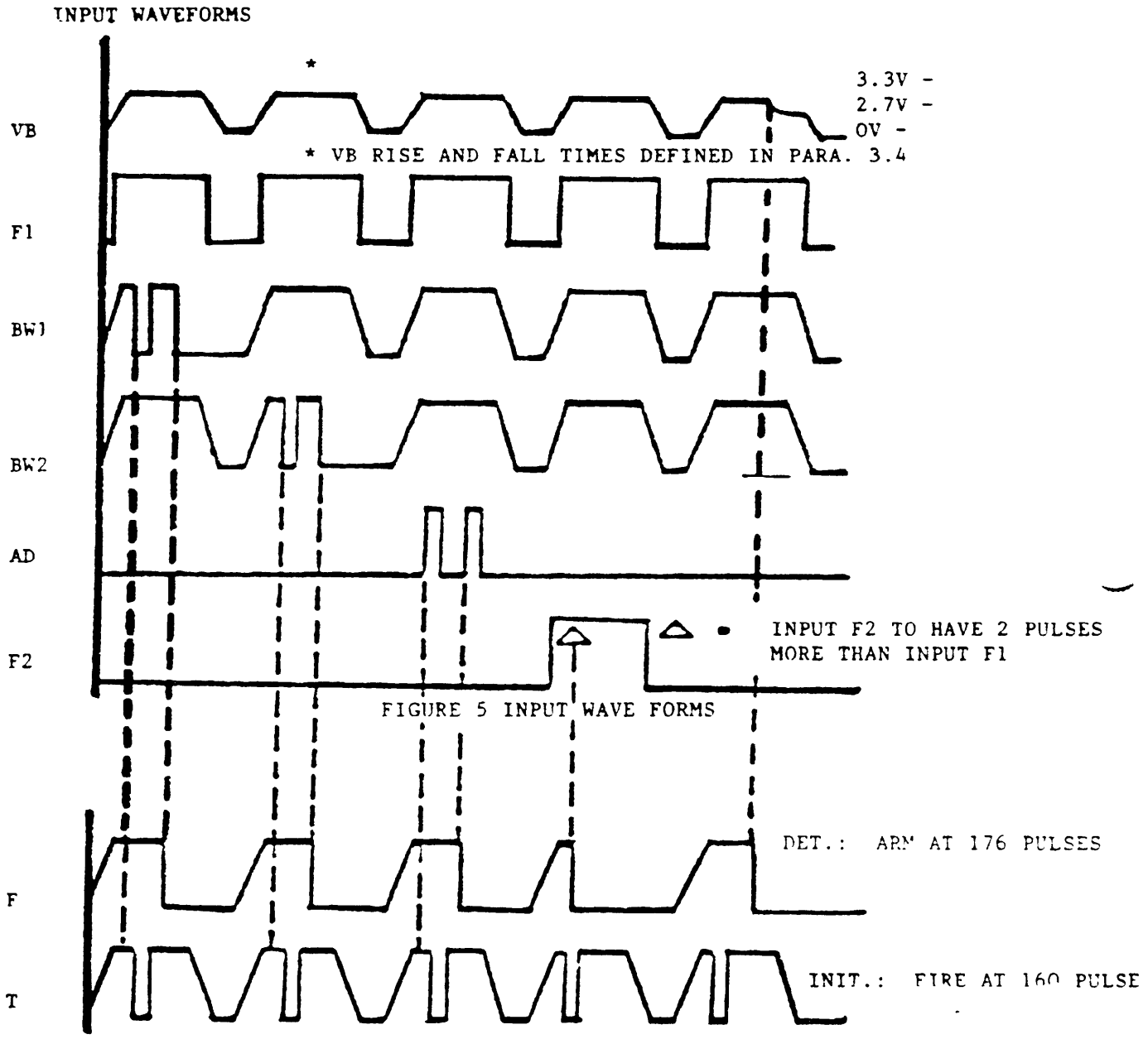


FIGURE 5 INPUT WAVE FORMS

FIGURE 6 OUTPUT WAVE FORMS

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40.2.5 L.V.D. level. With the S.U.T. connected as shown in Figure 7 and V_B=4.0V, a minimum of 76 pulses shall be applied on the F_1 input. The voltage on V_B shall be decreased at 100mv/SEC max until an output occurs on the F output. The voltage (VB) at which this occurs shall be recorded as the L.V.D. voltage.

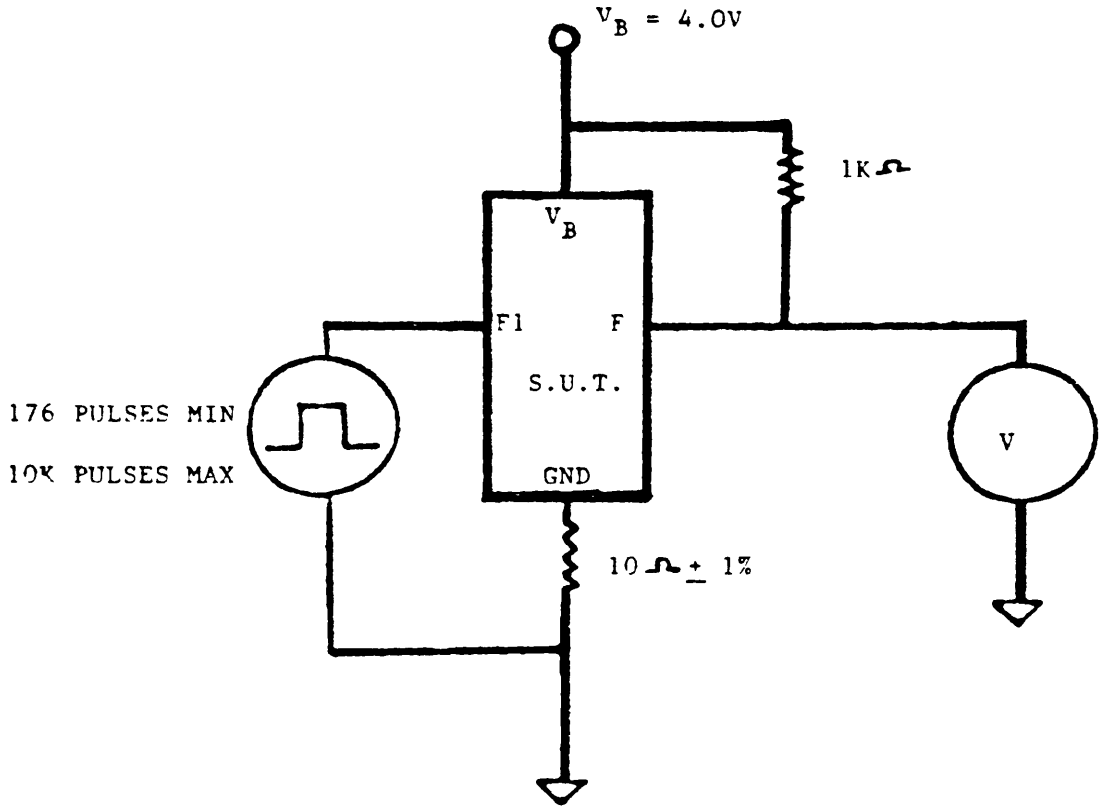


FIGURE 7. L.V.D. level measurement

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40.2.6 Output capacitor voltage. The voltage on the output capacitor shall be measured by connecting the S.U.T. as shown in Figure 8.

A. Capacitor present - with SW1 open apply power and verify that V_C rises to the limits specified in Table I. Take this measurement 1 second after power on.

B. Capacitor Leakage - Close SW1 until V_C = 3.3 volts, then open SW1. Upon opening SW1 wait 14 seconds minimum then measure V_C. V_C must be greater than the value specified in Table I.

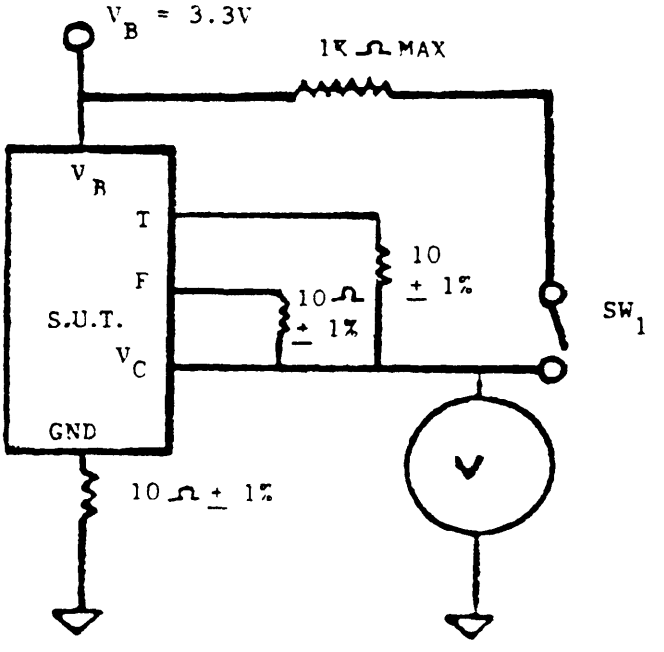
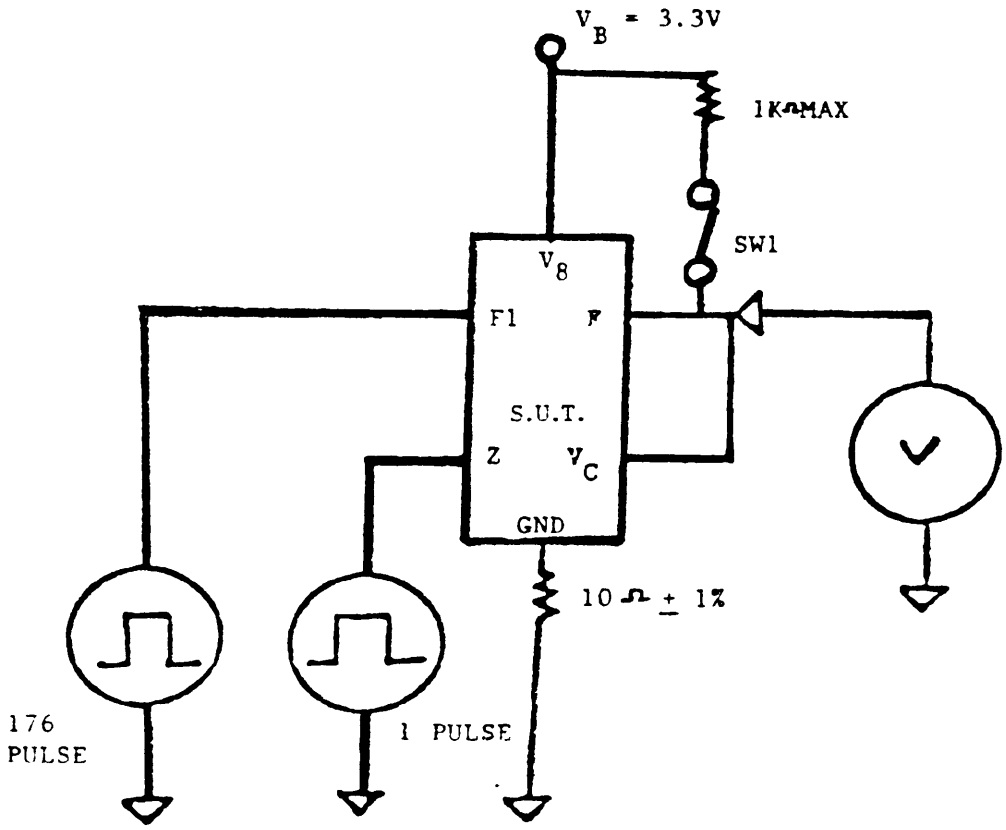


FIGURE 8. Output capacitor voltage measurement.

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40.2.7 Detonator firing circuit. the S.U.T. shall be connected as shown in Figure 9.* A minimum of 176 pulses shall be input on F_1 followed by 1 pulse input on Z. The output on P shall be measured and shall meet the requirements in Table I, after Z has an input.

* Quick charge the capacitor to VB by closing SW1 before the input of 176 pulses. Then open SW1 before giving an input to Z and measuring the output on F.



Input Z = any or all of BW1, BW2, AD.

FIGURE 9. Detonator output measurement.

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40.2.8 Gas generator firing circuit. The S.U.T. shall be connected as shown in Figure 10.* A total of 160 pulses shall be input on F_1. The output on T shall be measured and shall meet the requirements in Table I.

* Quick charge the output capacitor by closing SW1 before input of 160 pulses. Open SW1 before applying 160 pulses on F1.

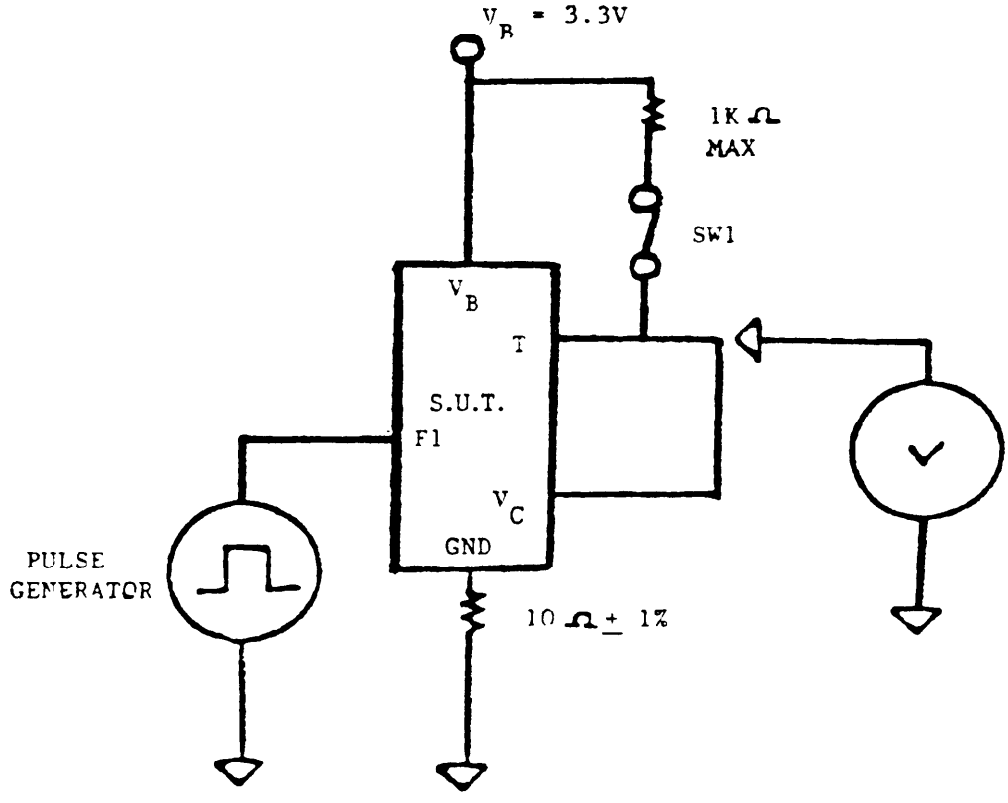


FIGURE 10. Gas generator output measurement.

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40.2.9 Detonator resistance. The detonator resistance shall be measured using a ohmmeter whose current output is 5.0 max. between F and S. The resistance measured shall be between the limits specified in Table I.

40.2.10 Gas generator resistance. The initiator resistance shall be measured using an ohmmeter whose current output is 5.0 ma max. between T and V_C (or preform). The resistance measured shall be between the limits specified in Table I.

40.2.11 Detonator short. The detonator short shall be measured using an ohmmeter whose current output is 5.0 ma max. between F and V_C (or preform). The resistance shall be within the limits specified in Table I.

40.2.12 Breakwire continuity. The breakwire continuity shall be measured by connecting the S.U.T. as shown in Figure 11. A high logical output must be present at both BW1 and BW2.

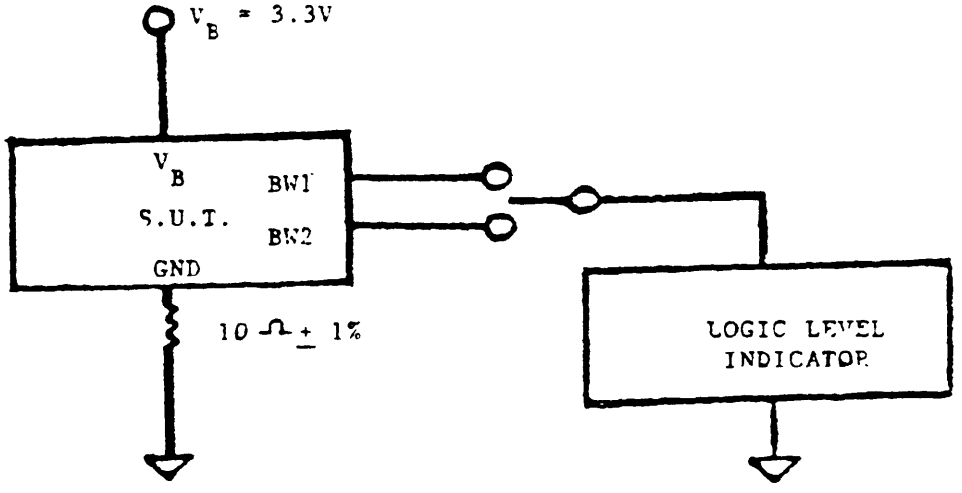


FIGURE 11. Breakwire continuity check.

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40.2.13 Initial current. The initial current shall be measured by connecting the S.U.T. as shown in Figure 12. The initial current shall be within the limits given in Table I, after a maximum stabilization time of 200 milliseconds.

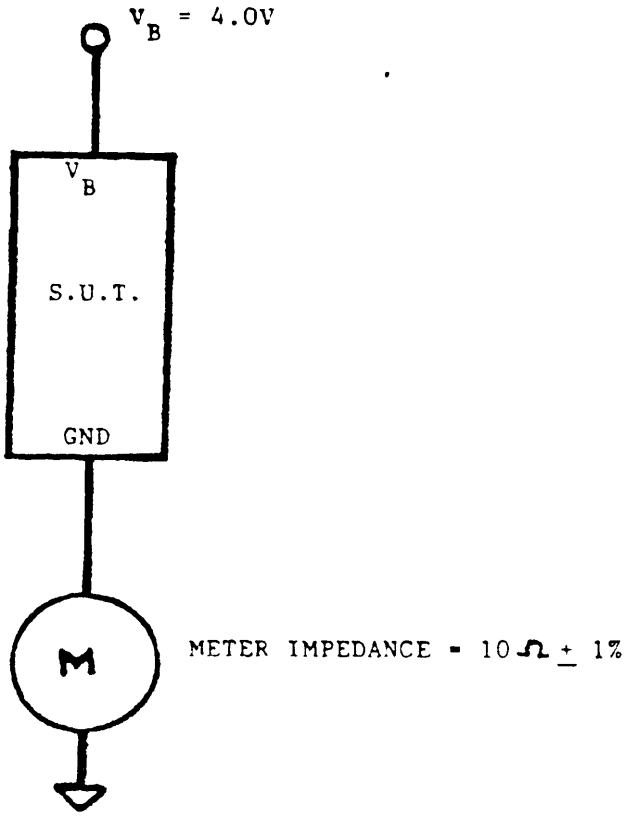


FIGURE 12. Initial current measurement.

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40.2.14 Final current. The final current shall be measured by connecting the S.U.T. as shown in Figure 13. After the inputs shown in Figure 14 occur, the output shall be within the limits given in Table I.

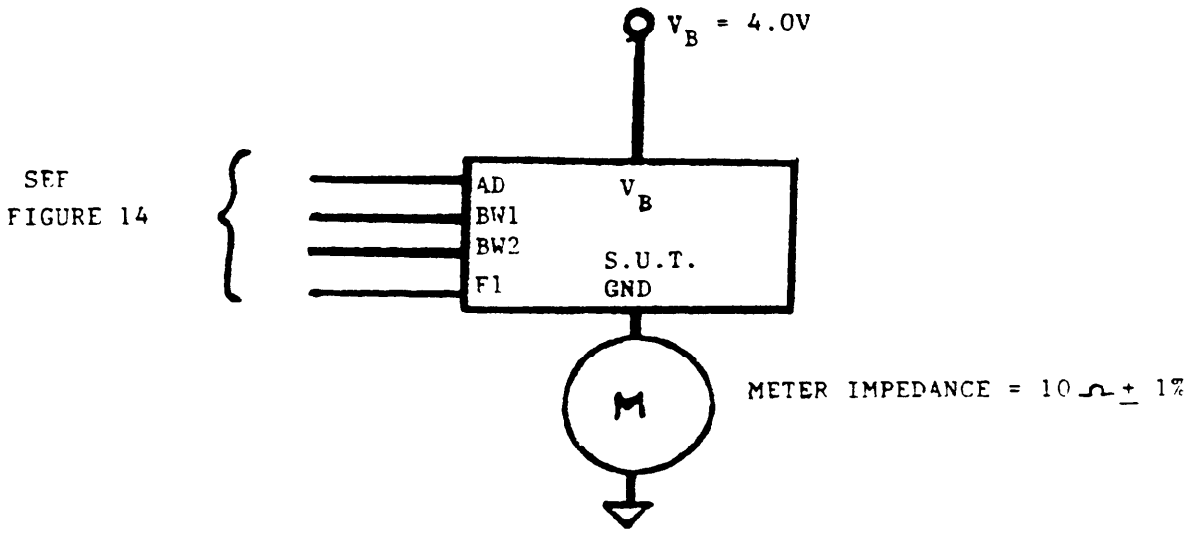


FIGURE 13. Final current measurement.

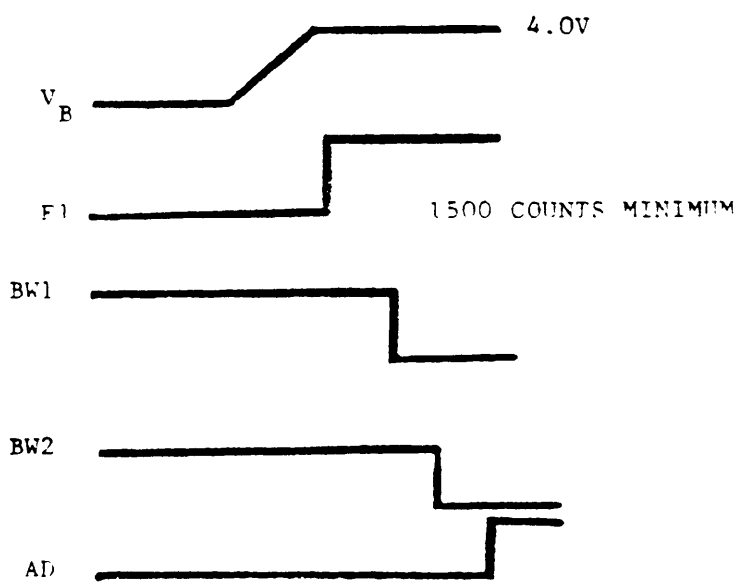


FIGURE 14. Waveform output for final current measurements

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40.2.15 AD switch operation. With the S.U.T. connected as shown in Figure 15, the S.U.T. shall be moved to verify AD switch function.

After the input of 180 pulses, the S.U.T. shall be moved thru an arc of 30° min. After movement is complete, an output logic "0" shall be measured.

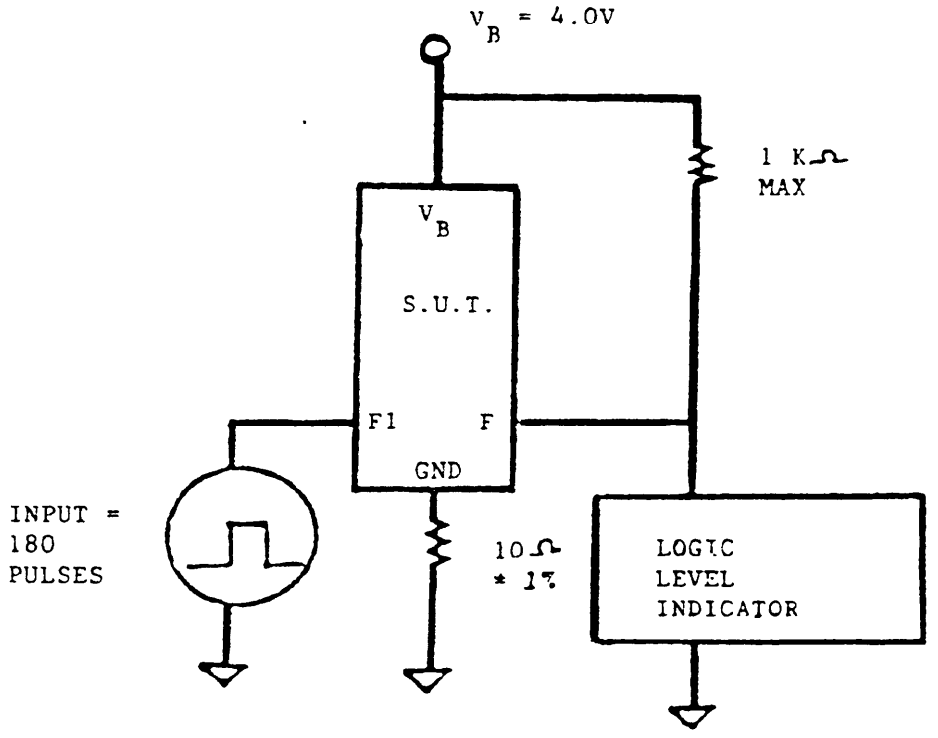


FIGURE 15. AD switch output measurement.

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